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IMPLEMENTATION COMPLETION AND RESULTS REPORT

<IDA 4764-BD; IDA 5736-BD; TF 15113-BD>

ON A

CREDIT AND GRANT

IN THE AMOUNT OF SDR123.6 MILLION AND CAD19.8 MILLION

(US\$195.7 MILLION EQUIVALENT)

TO THE

PEOPLE'S REPUBLIC OF BANGLADESH

FOR THE

SKILLS AND TRAINING ENHANCEMENT PROJECT

December 27, 2019

Education Global Practice
South Asia Region

CURRENCY EQUIVALENTS

(Exchange Rate Effective {Aug 06, 2019})

Currency Unit = BDT

BDT 84.495 = US\$1

US\$ 1.37623 = SDR 1

FISCAL YEAR

July 1 - June 30

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ABBREVIATIONS AND ACRONYMS

ADB	Asian Development Bank
ADM	Accountability and Decision-Making
ADP	Annual Development Plan
AF	Additional Financing
ASSET	Accelerating and Strengthening Skills for Economic Transformation
BGMEA	Bangladesh Garment Manufacturers and Exporters Association
BMET	Bureau of Manpower, Employment and Training
BTEB	Bangladesh Technical Education Board
CAS	Country Assistance Strategy
CFAA	Country Financial Accountability Assessment
CGA	Controller General Accounts
CPF	Country Partnership Framework
CPS	Country Partnership Strategy
DA	Designated Account
DO	Development Objective
DPP	Development Project Proposal
DTE	Department of Technical Education
EA	Environmental Assessment
EC	European Commission
EFA	Education for All
EIA	Environmental Impact Assessment
EMF	Environmental Management Framework
EMP	Environmental Management Policy
FM	Financial Management
FY	Fiscal Year
GDP	Gross Domestic Product
GOB	Government of Bangladesh
HRMIS	Human Resource Management Information System
HSC	Higher Secondary Certificate
HSC (Voc)	Higher Secondary Certificate (Vocational)
IBRD	International Bank for Reconstruction and Development
ICR	Implementation Completion and Results
ICT	Information and Communication Technology
IDA	International Development Association
IDG	Institutional Development Grant
IDP	Institutional Development Plan
IFC	International Finance Corporation
IFRs	Interim Financial Reports
ILO	International Labor Organization
IMC	Institutional Management Committee
IOI	Intermediate Outcome Indicator
IP	Implementation Progress
IPP	Indigenous Peoples Plan
ISC	Industry Skills Councils
ISR	Implementation Status Report
IUFR	Interim Unaudited Financial Report

KPI	Key Performance Indicators
MDTF	Multi-Donor Trust Fund
M&E	Monitoring and Evaluation
MOE	Ministry of Education
MOEWOE	Ministry of Expatriate Welfare and Overseas Employment
MOLE	Ministry of Labor and Employment
MOU	Memorandum of Understanding
MS	Moderately Satisfactory
MTR	Mid-Term Review
NER	Net Enrollment Rate
NYPi	Nanyang Polytechnic International
NSAPR	National Strategy for Accelerated Poverty Reduction
NSDP	National Skills Development Policy
NSDC	National Skills Development Council
NTVQF	National Technical and Vocational Qualification Framework
OI	Outcome Indicator
OP	Operational Policy
PAD	Project Appraisal Document
PD	Project Director
PDO	Project Development Objective
PDV	Present Discounted Value
PFM	Project Financial Management
PIC	Project Implementation Committee
PID	Project Information Document
PIM	Project Implementation Manual
PIU	Project Implementation Unit
PMT	Proxy Means Test
PR	Project Restructuring
PSC	Project Steering Committee
RADP	Revised Annual Development Plan
RF	Results Framework
RPL	Recognition of Prior Learning
SDR	Special Drawing Rights
SMF	Social Management Framework
SSC	Secondary School Certificate
SSC (Voc)	Secondary School Certificate (Vocational)
STEP	Skills and Training Enhancement Project
TF	Trust Fund
TOC	Theory of Change
TOR	Terms of Reference
TSCs	Technical Schools and Colleges
TTCs	Technical Training Centers
TVET	Technical and Vocational Education and Training
US\$	United States Dollars
WB	World Bank

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**DATA SHEET****BASIC INFORMATION****Product Information**

Project ID	Project Name
P090807	Bangladesh - Skills and Training Enhancement Project
Country	Financing Instrument
Bangladesh	Investment Project Financing
Original EA Category	Revised EA Category
Partial Assessment (B)	Partial Assessment (B)

Related Projects

Relationship	Project	Approval	Product Line
Supplement	P145118-Additional Financing: Skills and Training Enhancement Project	26-Sep-2013	Recipient Executed Activities
Additional Financing	P155389-Additional Financing Skills and Training Enhancement Project	30-Oct-2015	IBRD/IDA

Organizations

Borrower	Implementing Agency
PEOPLE'S REPUBLIC OF BANGLADESH	Directorate of Technical Education



Project Development Objective (PDO)

Original PDO

The Project Development Objective is to strengthen Training Institutions to improve training quality and employability of trainees, including those from disadvantaged socioeconomic backgrounds.

FINANCING

	Original Amount (US\$)	Revised Amount (US\$)	Actual Disbursed (US\$)
World Bank Financing			
P090807 IDA-47640	79,000,000	78,745,377	77,148,515
P090807 TF-15113	15,709,907	15,670,555	15,670,555
P090807 IDA-57360	100,000,000	100,000,000	97,263,909
Total	194,709,907	194,415,932	190,082,979
Non-World Bank Financing			
Borrower/Recipient	9,000,000	29,800,000	25,239,000
Total	9,000,000	29,800,000	25,239,000
Total Project Cost	203,709,907	224,215,932	215,321,979

KEY DATES

Project	Approval	Effectiveness	MTR Review	Original Closing	Actual Closing
P090807	10-Jun-2010	22-Aug-2010		30-Jun-2016	30-Jun-2019

RESTRUCTURING AND/OR ADDITIONAL FINANCING

Date(s)	Amount Disbursed (US\$M)	Key Revisions
18-Jan-2014	28.63	
21-Jan-2016	80.16	Additional Financing
29-Jun-2017	127.41	Change in Loan Closing Date(s)



KEY RATINGS

Outcome	Bank Performance	M&E Quality
Highly Satisfactory	Satisfactory	High

RATINGS OF PROJECT PERFORMANCE IN ISRs

No.	Date ISR Archived	DO Rating	IP Rating	Actual Disbursements (US\$M)
01	21-Feb-2011	Satisfactory	Satisfactory	4.93
02	21-Sep-2011	Satisfactory	Moderately Satisfactory	4.93
03	30-Apr-2012	Satisfactory	Moderately Satisfactory	10.61
04	29-Oct-2012	Satisfactory	Satisfactory	15.28
05	17-May-2013	Satisfactory	Satisfactory	20.06
06	06-Dec-2013	Satisfactory	Satisfactory	28.63
07	04-Jun-2014	Satisfactory	Satisfactory	34.60
08	30-Jul-2014	Satisfactory	Satisfactory	52.67
09	22-Jan-2015	Moderately Satisfactory	Moderately Satisfactory	52.67
10	31-Mar-2015	Satisfactory	Moderately Satisfactory	60.59
11	08-Oct-2015	Satisfactory	Moderately Satisfactory	75.21
12	26-Apr-2016	Satisfactory	Satisfactory	81.82
13	20-Sep-2016	Satisfactory	Satisfactory	109.29
14	22-Mar-2017	Satisfactory	Moderately Satisfactory	109.29
15	03-Oct-2017	Satisfactory	Moderately Satisfactory	127.41
16	03-Apr-2018	Satisfactory	Satisfactory	141.40
17	16-Nov-2018	Satisfactory	Satisfactory	159.71

**SECTORS AND THEMES****Sectors**

Major Sector/Sector	(%)
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Public Administration	5
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Other Public Administration	5
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Education	90
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Tertiary Education	90
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Industry, Trade and Services	5
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Other Industry, Trade and Services	5
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Themes

Major Theme/ Theme (Level 2)/ Theme (Level 3)	(%)
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Economic Policy	10
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Trade	10
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Trade Facilitation	10
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Private Sector Development	110
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Business Enabling Environment	10
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Innovation and Technology Policy	10
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Jobs	100
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Public Sector Management	5
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Public Administration	5
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Municipal Institution Building	5
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Human Development and Gender	76
Education	76
Access to Education	19
Science and Technology	19
Teachers	19
Standards, Curriculum and Textbooks	19

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I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES

A. CONTEXT AT APPRAISAL

Context

1. **At the time of appraisal of the Skills and Training Enhancement Project (STEP, the Project), Bangladesh had been enjoying continued economic growth, despite some slow-down due to the global economic environment, thanks to a vital contribution of the export sector and remittances from migrant workers.** The growth of RMG and knitwear exports had been impressive, and Bangladesh ranked second only to China in terms of global market shares. Exports accounted for an estimated 20 percent of GDP, making Bangladesh one of the leading exporting Least Developed Countries in the world. Bangladeshi migrant workers sent an average annual remittance of about US\$816 per annum that totaled an estimated US\$9 billion. However, the vast majority of Bangladeshi migrant workers were unskilled. Skilling the workforce could increase their wages in international labor markets and protect the workers from unemployment in the event of economic downturns. Further, about 85 percent of Bangladesh's exports were based on only seven or eight products, and continued weak demand from export markets was seen as have a potentially adverse impact on the country's GDP. The economy slowed down around 2008 due to the decline in aggregate demand, which had been affected by a decrease in exports and a reduction in the inflow of remittances as job losses mounted in host countries.
2. **The Government of Bangladesh (GOB) sought to boost economic growth by ensuring that its workforce was adequately skilled so that its key industries would remain competitive in global markets.** The size of the labor force had grown considerably, fueled by greater participation of women and youth. As a result, the demands on institutions delivering education and skills had increased significantly. Hence, the GOB sought to improve access to skills and training programs, improve their quality and effectiveness, and strengthen the overall governance in the education and skills sectors. Bangladesh had witnessed tremendous gains in primary and secondary education outcomes. Net enrollment rates (NER) at the primary and secondary levels were 90 percent and 45 percent, respectively, with gender parity. However, overall literacy rates were below 50 percent, and school completion rates were meager. The labor force's average years of schooling was 4.2 years, lower than that of the overall population's 4.4 years. Only a tiny proportion of the labor forces had any vocational education or training. Additional efforts were needed to harness the strength of its population.
3. **TVET provision in Bangladesh suffered from weak governance structures, low quality, and a lack of relevance of programs to the needs of the labor market, high rates of internal inefficiencies, and a lack of resources.** Access to TVET was low, especially for women and the economically disadvantaged youth. As a result, only a tiny proportion of the labor force had any vocation education or training. For those that did, the *quality of training* provided by TVET institutions was poor, with an overemphasis on theoretical work and minimal emphasis on practical work. Overly centralized structures and procedures made it difficult for institutions to ensure demand orientation and left little room for industries and employers to play a role in identifying the skills needed by industry. The TVET sector's *governance, management, and financing* made it challenging to improve the relevance and quality of training while ensuring access, with equity, to an ever-increasing number of skilled workers required by industry. The TVET sector is highly fragmented with more than 20 ministries offering programs with varied skills training programs. There was little or no coordination



or standards, poor standardization and equivalency among different programs and levels, weak accreditation to ensure that trainees have achieved the minimum level of expected competencies. Technical institutions had a large unutilized capacity, with low pass rates and employability of graduates. TVET financing was also low. Bangladesh spent less than 1.5 percent of its education budget on all technical education and training in the country.



Theory of Change (Results Chain)

PROJECT: PDO -- to strengthen Training Institutions to improve training quality and employability of trainees, including those from disadvantaged socio-economic backgrounds

Project Components	Activities	Outputs	Immediate Project Outcomes	Long-term Outcomes
Improve the Quality and Relevance of Training	- Support to Institutional Development Plan (IDP) of Public and Private Institutions Offering Diploma Programs - Support to IDP of Public and Private Short-term training Institutions - Provision of Stipends to disadvantaged student of diploma and short-term training institutions	- Institutional Development Grants (IDG) awarded competitively to Diploma Institutions - IDG awarded competitively to short-term training institutions - Stipends provided to disadvantaged and female students - Teachers contracted - Teachers trained	[PDO Indicators] - Pass rates of diploma programs increased - Employment rates and completion rates for trainees of short-term courses increased - Stipends increased and enrollment of disadvantaged and female students increased [Other Outcomes] - Enrollment in supported courses increased - Teacher vacancies reduced - Share of females in enrollment increased	- Enrollment in TVET increased - Private sector participates continuously as a partner in designing and delivering of TVET - Quality of TVET improved significantly - TVET contributes significantly to more and better employment of TVET graduates - All students interested in pursuing TVET are able to do so - TVET contributes significantly to meeting demand for skilled workers both in Bangladesh and abroad
Pilots In TVET	- Supporting Industry Skills Councils (ISCs) and the NSDC - Strengthening the Secondary School Certificate (Voc)	- ISCs/NSDC have office and TA and functioning - SSC (Voc) pilot apprenticeship program evaluated	Links between TVET sector and private sector strengthened	
Institutional Capacity Development	- Strengthening BTEB to support quality assurance and strengthen curriculum devt. - Improving BTEB monitoring and certification - Strengthening capacity for planning, curricula & standard setting for BMET - Implementation of Recognition of Prior Learning (RPL)	- Number of staff trained in core expertise areas in DTE, BTEB and BMET - RPL program become operational	- Increased sectoral capacity for improving quality, relevance and delivery of TVET programs, and coordinating delivery among sectoral institutions - Increased capacity for monitoring and evaluation, inter alia, through better and more timely information and results - Workers skills assessed through RPL	
Project Mgt, Communication and M&E	- Project Implementation Unit - Communications with stakeholders - Monitoring and evaluation of project activities	- Awareness raising campaigns of project-supported initiatives - Impact evaluation study	- More students become aware of TVET programs - Results of monitoring and evaluation are used as input to improve design and delivery of TVET programs	



Project Development Objectives (PDOs)

4. The Project Development Objective, as specified in the Financing Agreement, was to strengthen Training Institutions to improve training quality and employability of trainees, including those from disadvantaged socio-economic backgrounds. The wording of the PDO in the Project Appraisal Document (PAD) was slightly different and specified that the Project would strengthen “selected public and private training institutions”¹.

Key Expected Outcomes and Outcome Indicators

5. The key expected outcome was to make the TVET sector more responsive to current and emerging labor market needs for a skilled workforce. The PDO indicators described in the PAD were the following²:
 - Pass rates of students from supported Diploma programs;
 - Employment rates for trainees of short-term courses within 6 months of course completion;
 - Number of students receiving stipends in supported polytechnics and short-term training courses (student-years); and
 - Achievements in vocational competencies assessed on a sample basis.

Components

6. **Component 1: Improve the Quality and Relevance of Training (Total US\$72.9 million; IDA US\$65.4 million; Actual US\$155.7 million).** This component was envisaged to support public and private institutions chosen based on well-defined eligibility and selection criteria under two sub-components.
7. *Sub-Component 1.1: Window I: Support to Public and Private Institutions Offering Diploma Programs (Total US\$58.9 million; IDA US\$51.4 million)*³. Window I would focus on improving the quality and relevance of diploma level technical education in 33 public and private institutions based on eligibility and selection criteria. Once endorsed by the Project Implementation Committee (PIC), participating institutions would prepare a comprehensive Institutional Development Plan (IDP). Implementation grants would be subject to a ceiling of US\$1 million and financed (a) infrastructure rehabilitation and refurbishment; (b) modernization of equipment and teaching and learning instruments; (c) capacity development in the form of course/training materials, training, study tours, and exposure visits; (d) remuneration for guest faculty from industry; (e) teacher training; (f) students’ training materials and cost towards industry visits; and (g) improvements in the administration of institutions, outreach, communications, and placement improvements. An Annual Performance Grant of up to US\$200,000 would be awarded to the top five

¹ World Bank, Project Appraisal Document, Skills and Training Enhancement Project, Report No. 53549-BD dated May 11, 2010.

² The wording of the PDO Indicators in Annex 1 of the PAD differed slightly from that in the main text. The wording in Annex 1 is shown in this section as it was more detailed.

³ Bangladesh’s formal pre-employment Technical Vocational Education and Training (TVET) system consisted of three levels, with an intake capacity of about 500,000 students. The three levels include: (i) short-courses through a range of short-term, three to six-month training courses; (ii) two-year vocational programs for students in secondary and higher secondary education (SSC (Voc) and HSC (Voc)); and (iii) a four-year post-secondary Diploma program. Three agencies oversee the pre-employment TVET system: (i) the Directorate of Technical Education (DTE) under the MOE, responsible for setting the overall policy framework of vocational education and training; (ii) the Bangladesh Technical Education Board (BTEB), a statutory agency, also under the MOE, responsible for maintaining the training standards, conducting assessments and examinations, and providing student verification, and helping accredit institutions; and (iii) Bureau of Manpower, Employment and Training (BMET) under the MOEWOE focuses on estimating demand for labor in both domestic and overseas markets.



performing institutions, as measured at the end of the third year of implementation through performance indicators.

8. To increase access to post-secondary programs to students from socio-economically disadvantaged backgrounds, the Project provides stipends (US\$10/monthly)⁴ to students in eligible polytechnics who were estimated to be in the bottom two income quintiles. The screening was done using Proxy-Means Testing (PMT)⁵. Beneficiaries were expected to meet the following criteria to remain eligible: (a) maintaining 75 percent attendance, and (b) maintaining 45 percent pass marks in the annual examinations.
9. *Sub-Component 1.2: Window II: Support to Public and Private Institutions Offering Short-term Programs (Total US\$14.0 million; IDA US\$14.0 million).* Window II would support training programs of a minimum of 360 hours with a duration of six months or less for about 40-50 institutions. Institutions and courses offered would be selected based on development proposals submitted by the institutions, and those selected would be provided an implementation grant of up to US\$250 per student enrolled in the course. Proceeds of the grants covered (a) tuition costs to the institution; and (b) stipends of US\$10 per month to students in courses supported by the Project. The Project was expected to support over 70,500 trainees over its life.
10. **Component 2: Pilots in TVET (Total US\$5.5 million; IDA US\$5.3 million; Actual US\$6.2 million).** This component was designed to support two types of pilot activities aimed at strengthening the TVET system, through (a) direct start-up and operational support to Industry Skills Councils and the National Skills Development Council, and (b) assistance to Secondary School Certification (Vocational) schools (SSC (Voc)) to determine the most effective means to strengthen the programs.
11. *Sub-Component 2.1: Supporting Industry Skills Councils (ISCs) and the National Skills Development Corporation (NSDC) (Total US\$0.5 million; IDA US\$0.3 million).* This sub-component was designed to provide one-time grant financing and operational cost support during the five years of the program (on a declining basis) for the establishment of specific offices of about two to four ISCs or NSDC, in addition to all the outreach activities expected of the ISCs⁶. The Better Business Forum would also benefit from the support for the NSDC.
12. *Sub-Component 2.2: Strengthening the Secondary School Certificate (Vocational) (Total US\$5.0 million; IDA US\$5.0 million).* This sub-component was designed to implement and evaluate several different interventions and strategies to improve the effectiveness of essential vocational training opportunities for vulnerable students that were considered likely to take employment after secondary education rather than continuing further education. The existing SSC (Voc), as implemented, was deemed to be ineffective, with few students graduating with employable skills, and the pilots were expected to determine a more effective way of program delivery and improved employment outcomes.

⁴ The amount was roughly the minimum amount needed to cover out-of-pocket daily expenses for educational activities.

Bangladesh has a long history of providing stipend to boost school enrollment especially for girls. Those programs have been very popular among students and parents.

⁵ Using the same Proxy-Means Testing methodology as is used in Secondary Education Quality and Access Enhancement Project to determine the individual's poverty status

⁶ A number of ISCs were being established under the framework of the TVET Reforms Program of the EC-ILO with the aim of providing critical information to the TVET sector about the current and projected human resources and training needs in selected industries, accurate data on skills bottlenecks in these industries with the aim of helping to develop industry-led standards, and training products and services that will be recognized and accredited by national bodies such as BTEB, and by other international accreditation agencies, or by industry representatives. The NSDC role was to guide the overall regulatory and legislative framework for TVET and skills development in Bangladesh.



13. **Component 3: Institutional Capacity Development (Total US\$4.7 million; IDA US\$4.4 million; Actual US\$34.8 million).** This component was designed to strengthen the Government's capacity to manage the TVET sector by supporting institutional capacity development of the Directorate of Technical Education (DTE), the Bangladesh Technical Education Board (BTEB) under the Ministry of Education (MOE), and the Bureau of Manpower Employment and Training (BMET) under the Ministry of Expatriate Welfare and Overseas Employment (MOEWOE).
14. **Component 4: Project Management, Communications, and Monitoring and Evaluation (Total US\$4.9 million; IDA US\$3.9 million; Actual US\$13.0 million).** *Sub-Component 4.1: Project Management and Communications (Total US\$4.3 million; IDA US\$3.3 million).* This sub-component was designed to help establish a separate STEP Project Implementation Unit (PIU), a project management and implementation structure, to support the improvement of system management and implementation of reforms through training of policy planners, managers, and administrators and enhance communications.
15. *Sub-Component 4.2: Project Monitoring and Evaluation (Total US\$0.6 million; IDA US\$0.6 million).* This sub-component was designed to support the monitoring and evaluation (M&E) of project progress and outcomes, and enhancing the M&E capacities in the training institutions receiving support under the Project.

B. SIGNIFICANT CHANGES DURING IMPLEMENTATION (IF APPLICABLE)

16. **The STEP underwent two rounds of Additional Financing (AF) with project restructuring to meet emerging needs.** Each of the AFs and corresponding project restructurings incorporated lessons and findings from implementation and evaluations that ensured the Project's continued relevance to the Government's priorities. For instance, the teams identified issues with female enrollment that led to an adjustment to make stipends available to all female TVET students regardless of socio-economic standing⁷. Also, the Project was adjusted based on findings of the evaluation of the SSC (Voc) program that recommended downsizing the SSC (Voc) pilot towards an operational pilot to establish a standardized apprenticeship program that would strengthen the labor market linkages of SSC (Voc) institutions. The original Project financed the development of the Recognition of Prior Learning program, and based on progress, the subsequent AFs were adjusted to fund first, the roll-out of RPL to several assessment centers and implementation, and later, an impact evaluation of the program. Also, given the Government's plans to promote efficiency, transparency, and accountability in public financing by moving to e-Government procurement, the second AF responded to its request to move to e-Government Procurement in central TVET agencies and selected polytechnics.
17. The first AF in the amount of US\$16.71 million, was approved in September 2013 under a Multi-Donor Trust Fund with funding from Canada. At the time, US\$2.75 million had been disbursed from the original credit, and the Project was rated Moderately Satisfactory (MS) for both Development Objectives (DO) and Implementation Performance (IP). The AF was to help finance the costs associated with the need to provide additional resources under the original components of the Project to (a) scale-up ongoing institutional grants to additional public and private institutions; (b) fill the funding gap for stipends which emerged as a result of higher demand for the poverty-targeted stipends than originally anticipated; and (c) scale-up

⁷ It was recognized that most of female students were screened out in the means-testing process for stipend. Female students in post-secondary TVET tend to be from less impoverished families compared male students as only families with some financial means would consider post-secondary education for girls. To address this reality and achieve the goal of increasing female enrollment, stipends were made available to all female students in the supported polytechnics.



capacity building, studies, communications and monitoring, and evaluation activities. For sub-component 2.2: Strengthening the SSC (Voc), it was decided to finance a pilot for the apprenticeship program. The pilot would address the disconnect between SSC (Voc) programs with employers' skill needs by establishing an institutionalized apprenticeship program.

18. A second AF in the amount of SDR 71.3 million (US\$100 million equivalent) was approved in October 2015. At the time, US\$19.5 million had been disbursed, and the Project was rated MS for both DO and IP. The AF was to help finance the costs to (a) scale-up ongoing institutional grants to additional public and private institutions; (b) fine-tune existing schemes and mechanisms, including (i) refinement of stipend program; (ii) greater flexibility in technology/trade support in polytechnics and short-courses; (iii) introduction of e-government procurement (e-GP); and (iv) enhancement of communication and mobilization activities; and (c) introduce further innovations: (i) digitizing teaching and learning contents; (ii) improving in-service pedagogical and subject training in Technical Teachers' Training College (TTTC), Technical Teachers' Training Institute (TTTI) and Vocational Teachers' Training Institute (VTTI); and (iii) impact evaluation for RPL activities. The RF was revised to reflect the expanded scope; and the introduction of sub-indicators for disaggregated groups (gender and disadvantaged). The Closing Date of the AF was set on June 30, 2019.

Revised PDOs and Outcome Targets

19. Revisions to the PDO Indicators in the AFs and PRs are presented in Table 1. Revisions to Intermediate Outcome Indicators are presented in Annex 1.
20. In addition, the Intermediate Results Indicators were adjusted to reflect the following: (a) new targets to cover the Project's expanded scope; (b) a minor rewording of indicators; (c) deleting the intermediate outcome indicator related to "internal revenue generated as a share of recurrent expenditure" as it would require legislative changes; and (d) setting targets for two intermediate outcome indicators, "evaluation of SSC (Voc) pilot" and "impact evaluation".

Table 1. Revision of PDO Indicators

No.	PDO Indicators	Changes made and reasons	EOP Targets		
			Original	1 st AF	2 nd AF
1.	Pass Rates of students from supported Diploma Programs	New targets were added due to the expanded project scope and duration.	60	60	65
	Female (sub-indicator)	This sub-indicator was added to track the outcome for disadvantaged groups	-	-	68
	Disadvantaged Students (sub-indicator)	This sub-indicator was added to track the outcome for disadvantaged groups. Disadvantaged students refer to students receiving stipends.	-	-	60
2.	Employment Rates for trainees of short-term courses within 6 months of course completion	New targets were added due to the expanded project scope and duration.	44	44	46
3.	Number of Students receiving stipends in supported	Modified to "No. of Disadvantaged students supported through stipends selected based on household information based on	32,000	97,000	175,000



	Polytechnics and short-term training courses (student-years)	selection criteria” to reflect the total number of disadvantaged students ^(*) .			
4.	Achievements in vocational competencies assessed on a sample basis	Modified to intermediate results indicator as the nature of the activity is piloting a new assessment concept and is part of the larger system improvement rather than an outcome by itself.	TBD	-	-

Note: (1) Students enrolled in the eligible short-term training courses were automatically qualified to receive stipends.

Revised Components

21. The first AF adjusted Sub-component 2.2, Strengthening the SSC (Voc), to finance an operational pilot that would respond to one of the key findings of the assessment study that was discussed during the Mid-Term Review. The operational pilot would address the disconnect between SSC (Voc) programs with employers’ skill needs by establishing an enhanced apprenticeship program that would strengthen the labor market linkages of SSC (Voc) institutions by mobilizing various stakeholders and preparing operational manuals.

Other Changes

22. The process of awarding Annual Performance Grants to IDGs under Sub-component 1.1 was canceled. This cancellation was because institutions faced initial challenges in the utilization of grants due to limited absorptive capacity. This dampened the motivation of awarding additional grants to the institutions.
23. The first AF included a reallocation of resources from Component 2 to Component 1 to fill the funding gap under the stipend programs due to higher than estimated demand from project beneficiaries. It also supported the decision to extend stipends to all female students in supported polytechnics to increase their participation and decrease their drop-out rates. The second AF also adjusted project costs and the implementation schedule, procurement arrangements (to introduce e-GP and framework agreements). The Closing Dates for both the original IDA Credit and the MDTF were extended from June 30, 2016 to June 30, 2017 to align better with the AF closing date of June 30, 2019.

Rationale for Changes and Their Implication on the Original Theory of Change

24. The changes described above were mostly to reflect new scope and targets resulting from AFs and the expansion of activities given implementation progress. The modifications caused no change in the Project’s Original Theory of Change, and only intensified efforts to achieve expected outcomes. The expansion of stipends to all female students strengthened the Project’s goal of increasing female enrollment in TVET as it gave a firm assurance and strong incentive of financial support to families of female students. In the country context, female students who were enrolling in TVET institutions tended to be from non-poor families that



could afford post-secondary education. As a result, the design of the stipend program provided solely based on means-testing results was largely excluding female students.

II. OUTCOME

A. RELEVANCE OF PDOs

Assessment of Relevance of PDOs and Rating

25. **The Relevance of the PDOs against the World Bank's Country Partnership Framework for Bangladesh for the Period FY16-FY20 is High⁸.** At Appraisal, the Project's objectives and design were relevant to the development priorities of the Bangladesh government and consistent with the World Bank's CAS to address the skills needs and gaps to enhance the competitiveness of Bangladesh in the global market and boost export-led growth. The government was keen on increasing the skills level of the entire population to increase the return to the economy and remittance from migrant workers.
26. Skills development continues to be the priority for the Bangladesh government and the World Bank's engagement. At closing, the PDO remains well aligned with the strategies outlined in the World Bank Country Partnership Framework for Bangladesh for the Period FY16-FY20, specifically under its Focus Area 2: Social Inclusion. The CPF highlights that Bangladesh's sustained growth will require continued attention to worker productivity and the protection of the poor. WBG activities will aim to consolidate the equity and access gains in health and education with constant due regard to gender and address the next generation of challenges related to quality improvements in health, access to clean drinking water and sanitation, and education and skills development. The Project is in full alignment with this country strategy of the WBG. In the labor market, most of the workforce remain undereducated, and the skills mismatch poses a significant recruitment problem for employers, putting worker productivity and the protection of the poor at risk. In the skills sector, the Project supported efforts to reform technical and vocational skills development to forge stronger links with industry and reduce skills mismatch.

B. ACHIEVEMENT OF PDOs (EFFICACY)

27. The Project Development Objective, as specified in the Financing Agreement, was to strengthen Training Institutions to improve training quality and employability of trainees, including those from disadvantaged socio-economic backgrounds. **The PDO had the following three objectives. (1) To strengthen training institutions to improve quality. (2) To strengthen training institutions to improve the employability of trainees. (3) To improve training quality and employability of trainees from disadvantaged socio-economic backgrounds.** The Project's PDO remained unchanged throughout implementation. Minor adjustments in its RF

⁸ World Bank, Country Partnership Framework for Bangladesh for the Period FY16-FY20, Report No. 103723-BD dated March 8, 2016.



were to reflect new targets resulting from AFs and the expansion of original activities and the introduction of new ones based on the progress in TVET reform actions. Therefore, this analysis does not evaluate the Project efficacy weighted by disbursements in after each of AFs/PRs. The Project was never at risk, as reflected in the Implementation Performance and PDO ratings in its ISRs. Split ratings should not alter the overall efficacy rating. Details on project outputs are in Annex 1.

28. **The Project was hugely successful in the achievement of its objectives.** At completion, STEP had exceeded by far targets for all its Outcome Indicators and all but one of its Intermediate Outcome indicators. More importantly, the Project's catalytic effect of its various activities has resulted in profound changes in Bangladesh's TVET sector. There has been a marked transformation in the public's perception of technical education. An ever-greater number of aspiring students see technical education as an attractive option⁹. Participating institutions have witnessed the benefits of establishing strong linkages with industry¹⁰. Partnerships forged with foreign technical training institutes have raised the bar in terms of expectations for training quality.

Assessment of Achievement of Each Objective/Outcome

29. **Objective 1 – Strengthen training institutions to improve quality:** Progress towards achievement of this objective was **High**. This objective was measured by the PDO Indicator 1, which tracked the pass rates of students from supported diploma programs. **There was notable achievement towards this objective, exceeding expectations as measured by the key indicators to measure progress.** The pass rate¹¹ increased from 50 percent at baseline to 86.5 percent, exceeding the revised target of 65 percent. It is also above the national average pass rate of 77.9 percent. In 2016, the government initiated a reform in examination methodologies, which resulted in fluctuation in the pass rates in the next two years¹². However, the project supported polytechnics demonstrated continuous improvement trajectory in their pass rates and also marked significantly higher pass rates than the national average¹³.

⁹ At the baseline, supported polytechnics enrolled only 55 percent of the sanctioned intake capacity. By the project completion, 99 percent of the intake capacity was filled, while the number of intake capacity also increased significantly.

¹⁰ All of the project supported institutions – both polytechnics and short-course providers – have established Institution Management Committee (IMC) which includes representatives from local industries to involve industries and employers in management and teaching processes.

¹¹ Pass rate refers to the rate of students passing the final national examination at the end of four year diploma course. This examination determines whether students are qualified to graduate from the course. Transition rate, which is used as an Intermediate Results Indicator to measure student retention, refers to the rate of students passing the year-end examination which determines whether students can move on to the next grade.

¹² For instance, in 2016, overall pass rate shot up significantly. From 2017 onwards, pass rates stabilized as the transition was completed. As such, the use of pass rate as a proxy of quality continues to be valid.

¹³ It should be noted that the Project's supported institutions were selected based on their merit and are likely to have had better pass rates than others. Nevertheless, overall improving trend for supported institutions demonstrate the steady improvement over time, and the substantial gap from the national average attests to significant gains in quality.



30. The performance of both females and disadvantaged students exceeded that of the average of all students: the overall pass rates of females and disadvantaged students were 89.5 percent and 92.5 percent, respectively.
31. The IDG activities, such as strengthening linkages to the industry, led to the improved learning experience and increased popularity of technical education at polytechnics and short-course institutions. The provision of stipends to disadvantaged students also had positive impacts on the learning experience. The shortage of teachers was a long-standing concern in polytechnics institutes. By contracting and training over 1,100 contractual teachers, the share of vacancies in sanctioned faculty posts of project-supported courses fell from 25 percent at baseline to 3 percent at completion (against a target of 5 percent).
32. Pedagogical and subject-based training of teachers received special attention under the Project. A total of 7,980 teachers of polytechnics and short-term courses received training against a target of 6,000. The Project established the Dhaka Technical Teachers Training Institute, which trained 1076 and 1126 teachers, respectively, for subjects and pedagogy. It formed a formal partnership with Nanyang Polytechnic International (NYPi) in Singapore to train 1,794 teachers and 158 managers of polytechnics. It also went into a formal partnership with Guangzhou Industry and Trade Technician College in China to train 553 teachers and 44 managers of short-course institutions. A teacher training survey conducted in November 2015 in nine project-supported polytechnics found that, overall, teachers were satisfied with the training.
33. STEP also supported a range of awareness-raising regarding technical education through posters, radio and television clips, the internet, and other means. The determination to improve the quality of training coupled with initial positive results have sparked a renewed interest in technical and vocational education. By completion, 36 awareness-raising campaigns had been carried out against an end-of-project target of 30 campaigns. Enrollment as a proportion of sanctioned intake capacity in supported diploma courses increased from 55 percent at baseline to 99 percent at completion (against a target of 95 percent).
34. The academic performance of students has improved and resulted in better student retention. The transition rate from the second to the third semester in supported diploma courses increased from 80 percent at baseline to 98 percent at completion (against a target of 95 percent). The completion rates in supported short-courses also increased significantly from 50 percent at baseline to 96 percent at completion (against a target of 95 percent).
35. The Satisfaction Survey (student, faculty, and employers) for polytechnic institutions provides highly encouraging results that support the above findings.¹⁴ Both students and teachers of beneficiary institutions rated significantly higher than those of the control group on most of the indicators of quality improvement, as summarized in Table 2. More importantly, the responses of the majority of the employers interviewed in the survey are encouraging. Twenty-one percent of employers maintain regular contact with polytechnics. Fifty-nine

¹⁴ The Satisfaction Survey (Student, Faculty and Employers) for IDG recipient institutions under the Skills and Training Enhancement Project (STEP) dated December 2, 2018 surveyed 4975 students and 482 teachers of 33 polytechnic institutes (of which 18 were public and 5 were private STEP beneficiaries and 8 were public and 2 were private control) and 66 public and private employers with a structured questionnaire and held 27 focus group discussions in 9 polytechnics with faculty members.



percent reported that there was a good demand for graduates from these polytechnics in the job market. Eighty-two percent found that these graduates were good workers with the desired skills. The only area they found weak was the effectiveness of the job placement cell. Overall, three-quarters of the employers were satisfied with the graduates of project-supported polytechnics whom they hired over the previous 12 months.

Table 2: Average Rating of Students and Teachers on Selected Indicators of Quality Improvement

Topic	Indicator	Response of Students		Response of Teachers	
		Beneficiary Polytechnic	Control Polytechnic	Beneficiary Polytechnic	Control Polytechnic
Curricula, Textbooks	Standard of textbooks	7.4	7.17	6.85	6.30
	Relevance of project work	8.0	7.58	8.63	8.07
Quality of Teaching	Theoretical class	7.67	7.18	8.09	7.74
	Practical	7.13	6.22	7.22	7.11
Teaching Facilities	Laboratory equipment	6.61	4.91	7.37	6.41
	Computer/software	6.39	4.83	-	-
Teachers' Capacity	Teaching complex subjects	7.64	7.26	8.56	8.44
	Overall teaching capacity	8.00	7.67	8.63	8.22
Career Development	Assistance by the polytechnic	7.39	6.73	8.12	8.01
	Effectiveness of placement cell	6.05	5.07	7.04	6.59

Note: Ratings are on a scale of 10, where 1 is least satisfied, and 10 is the most satisfied.

36. Objective 2 – Strengthen training institutions to improve the employability of trainees:

Progress towards achievement of this objective was **High**. The success of this objective was measured by the employment rate for trainees of short-term courses within six months of course completion¹⁵. The employment rates for trainees of short-courses within six months of course completion increased from 30 percent at baseline to 51 percent at completion, against a formally revised target of 46 percent. This improvement in employability is particularly remarkable in the job market context, where youth unemployment continued to be a thorny challenge. The Labour Force Survey (LSF) 2010 reported 5.4 percent unemployment rate for the age group of 25-29 years, while the LFS 2017 reported 5.7 percent unemployment rate for the age group of 25-34 years, which shows no improvement in job market environment for youths who are the majority of short-course trainees. There was also a notable achievement in strengthening training institutions to improve the employability of trainees, as measured by the key indicators selected to measure progress.

37. Both polytechnics and short-course institutions implemented practical actions to improve the employability of trainees. All the beneficiary institutions have established **Institutional Management Committees (IMCs)**. The IMCs are composed of representatives of the training institution and local industries and ensure that the training meets the needs of local industries. Beneficiary polytechnics have established over 300 **formal partnerships with industries** to

¹⁵ The Project focused on the employment rate of short-course graduates, rather than that of polytechnic graduates, as the post-graduation progression of polytechnic students is diverse. It is often the case that polytechnic graduates move on to higher education or other further education opportunities, depending on the availability of such opportunities. This situation makes the use of the employment rate of polytechnic graduates less reliable.



cement strong institution-industry linkages. These partnerships have promoted the refurbishment of labs and facilities, industry exposure for students through seminars by guest lecturers, and support for job placement. STEP also promoted the establishment of **job placement cells** in all project-supported polytechnics and short-course institutions. The mandate of these offices is to provide career counseling and job search skills training, facilitate the transition to employment, and help intermediate job placement between graduates and potential employers. The Project organized 33 **job fairs** where job seekers can contact interested employers. DTE has implemented a **job information portal** where graduates can find information on employment opportunities. The success of the first job fair, where around 1,200 students received job offers or commitments from employers, leads to the carrying out of this event yearly. The **National Skills Competition** was started in 2014 and held each year successfully to help showcase the skills, talent, and potential for innovation of polytechnic students and promote the value of technical training.

38. **The pilot apprenticeship program provided workplace experience to 9,125 SSC (Voc) students to enhance their practical skills and employability.** The apprenticeship program was offered in five trades (electrical, refrigeration and air conditioning, welding, dressmaking and tailoring, and food processing) in 20 institutes (10 public and 10 private). The apprenticeship program aimed to enhance the job relevance of the SSC (Voc) program through the partnership with industry. Upon completion, students were assessed under the NTVQF assessment framework. The ex-post assessment study of the SSC (Voc) Apprenticeship Program found that 62 percent and 25 percent of students in the pilot program were, respectively, highly satisfied and satisfied with the quality of the program¹⁶. Eighty-six percent of the beneficiary students reported that participation in the pilot program increased their skills. Eighty percent of them found that industry training increased their self-confidence. Because of their age, only 5.0 percent of pilot program students in the treatment group sought employment upon completion. Most of the other students are still in the education system. The average monthly income of these employed graduates was about 250 percent above that of employed graduates in the control group.

39. **To improve job market opportunities for informal sector workers, STEP piloted and expanded Recognition of Prior Learning (RPL).** The STEP rolled out the RPL program initially to 10 assessment centers affiliated to BTEB, and then to an additional 10 centers under the second AF. In all, 39,060 persons enrolled in the program to be assessed, exceeding the end-of-project target of 30,000 persons. The assessment of the RPL found that participants found the orientation program useful, relevant to the practical field of work, with suitable raw materials and tools for the RPL assessment.¹⁷ The assessment reports that there was a significant increase in full-time and part-time wage employment in public and private entities after the RPL assessment, while there was a substantial decrease in self-employment,

¹⁶ Evaluation of SSC (Voc) Apprenticeship Piloting Program dated July, 2018 surveyed 700 students in 7 SSC (Voc) institutes that participated in the pilot (treatment group) and 400 students in 7 SSC (Voc) institutes that did not participate (control group) through face-to-face interviews using a structured questionnaire. Focus group discussions were held with teachers of SSC (Voc) institutes participating in the pilot, and key informant interviews were held with directors/principals/heads of those institutes, with trainers and owners of industry partners involved with the industry training program and with employers of pilot program graduates.

¹⁷ Report on Recognition of Prior Learning (RPL) Assessment, dated August 2016. The assessment collected data from 600 Competent, 200 Partially Competent assesses, from 10 RPL assessment centers and 200 control respondents (who applied but were not selected for RPL assessment, through face-to-face interviews using a structured questionnaire. Focus group discussions were held with teachers of RPL assessment centers, and key informant interviews were held with directors/principals/heads of assessment centers, BTEB registered industry assessor, BTEB officials and employers of the RPL assesses.



unemployment, and student/trainees. Most of the Competent (77 percent) and Partially Competent (91 percent) assesses reported greater social recognition. More importantly, the monthly income of RPL participants was significantly higher for those that received Competent and Partially Competent assessments alike, as presented in Table 3.

Table 3: Minimum, maximum and average monthly income of RPL assessees

	Minimum Salary BDT		Maximum Salary BDT		Average Salary BDT	
	Competent	Partially Competent	Competent	Partially Competent	Competent	Partially Competent
<i>Wage employed</i>						
Before Assessment	1,500	1,500	18,000	17,000	6,226	6,163
After Assessment	3,000	2,500	30,000	28,000	13,228	11,967
<i>Self-employed</i>						
Before Assessment	9,000	9,000	25,000	25,000	14,183	12,170
After Assessment	15,000	15,000	45,000	40,000	18,983	15,990

40. **The Project also addressed the system-level capacity to enhance the market relevance of TVET programs through its support to ISCs.** There are 12 ISCs in Bangladesh as of December 2019. The ISCs have an institutional mandate to create linkages between skills program providers and private industries. Development of competency standards, participation in curriculum revisions, labor market analysis, and coordination of skills training are part of their vital missions. Different development partners support ISCs. The Project took up two ISCs – ICT ISC and Light Engineering – to provide support through NSDC. Although there were challenges in coordination and institutional capacity of NSDC and ISCs, the Project supported ICT ISC to develop market-aligned competency standards in several ICT occupations and integrate them into the national framework.
41. **Objective 3 – Improve training quality and employability for trainees from disadvantaged socio-economic backgrounds.** Achievement of the inclusiveness aspect of this objective was measured by the number of disadvantaged students supported through stipends selected based on household information-based selection criteria (PDO Indicator 3). The sub-indicators (female students and low-income students) of PDO Indicator 1 measured the quality aspect of this objective, while the gender breakdown of PDO Indicator 2 measured the employability aspect. The employability of low-income students was not adequately measured; therefore, progress towards achievement of this objective was **Substantial**.
42. The Project successfully contributed to enhancing the inclusiveness of TVET. Stipends supported a total of 285,422 disadvantaged students in supported polytechnics (53 public and 109 private), against the revised target of 175,000. Stipends also supported 139,477 students of short-courses. The Project also succeeded in increasing female enrollment in short-courses from 5 percent at baseline to 32 percent at completion, exceeding the revised target of 30 percent.
43. Disadvantaged students performed well in the academic outcome, indicating improved training quality for them. Stipend recipients outperformed the average pass rates: 89.5



percent for females and 92.5 percent for low-income students against an average of 86.5 for all students.

44. A study was carried out to evaluate the effectiveness of the stipends¹⁸. It was found that the Project had been successful in increasing enrollment and increasing transition rates. Application to public polytechnics became more competitive; the acceptance rate dropped from 37 percent in 2010 to 26 percent in 2013. Academic ability of applicants to public polytechnics witnessed steady improvement from 2010 to 2013 with a gradual increase in the proportion of applicants with an A+ grade. This trend was more pronounced for female applicants. More importantly, the study found that students receiving stipends on average outperformed those that did not by 0.04 grade point average points.
45. The Project expanded enrollment and improved the employability of female short-course trainees. The PDO Indicator that measured the employability of female students is the gender breakdown of PDO Indicator 2. The Project's six-monthly monitoring reports monitored this indicator by gender. They showed that the Project exceeded expectations as the female employment rate for short course graduates reached 58 percent against the target of 46 percent by completion.
46. Short-courses under the Project expanded access to skills training opportunities for low-income students and produced good employment outcomes. A tracer study of short-term training graduates found that the Project was successful in attracting poorer students to the supported short-courses through stipends and free tuition¹⁹. It also found that the employment rate among Project-supported graduates was higher than that of non-supported graduates from the same institutions. One of the critical factors was job placement support. Although weak in most institutions, it was better catered and utilized by Project-supported students.

Table 4: Summary of Objectives and Achievement of PDO Indicator

Unpacked Objectives	PDO Indicators	EOP Targets	Achieved
1. To improve training quality for trainees	Pass rates of students from supported Diploma programs (%)	65	86.5
2. To improve the employability of trainees	Employment rates for trainees of short-term courses within 6 months of course completion	46	51
3. To improve training quality and employability for trainees from a disadvantaged socio-economic background	Number of disadvantaged students supported through stipends selected based on household information based selection criteria	175,000	240,118

¹⁸ Ahmed, Shaikh S., Stipend Effectiveness of the Skills and Training Education Project (STEP), July, 2015.

¹⁹ World Bank, South Asia Region Education Global Practice, How Does the Short-Term Training Program Contribute to Skills Development in Bangladesh? A Tracer Study of the Short-Term Training Graduates, September 2015.



	Pass rates of students from supported Diploma programs (%) <u>for Female</u>	68	89.5
	Pass rates of students from supported Diploma programs (%) <u>for Disadvantaged</u>	60	92.5
	Employment rates for trainees of short-term courses within 6 months of course completion <u>for Female</u> ^{*1}	46	58

Note: (*1) the results framework did not have gender disaggregation for this indicator; gender-disaggregated employment records were provided in the project's six-monthly monitoring reports.

Justification of Overall Efficacy Rating

47. Overall achievement of objectives was high based on the achievement of the PDO as measured by the PDO Indicators, Intermediate Indicators, study findings, M&E reports, and other evidence. Considering that the outcome of its Objective 3 was not fully measured by PDO Indicators, Overall Efficacy is rated **Substantial**.

C. EFFICIENCY

48. This section presents the Project's efficiency in both economic terms and also in terms of the efficiency with which it was implemented. Based on both of these, the Project's efficiency is rated High. The updated economic analysis is summarized below. It is described in detail, together with the input parameters used for the analysis, in Annex 4.
49. Following the analysis prepared at Appraisal of the second AF, the economic impact of STEP is estimated as the increased lifetime labor earnings of the Project's beneficiaries. The Project's three significant benefits are generated from the following beneficiaries: (a) polytechnic graduates who would not have graduated in the absence of the Project's institutional grants for improving the quality of training (Sub-component 1.1); (b) graduates of polytechnics who would not have graduated in the absence of stipends provided by the Project (Sub-component 1.1); and (c) the employed graduates of short-term programs (Sub-component 1.2). The PDV of expected economic costs comprises both direct costs (the Project's cost and students' out-of-pocket costs) and indirect costs (students' forgone labor earnings) of completing the TVET diploma.
50. The ex-post cost-benefit analysis is evaluated using the outcome indicators (real pass rates and the number of polytechnic graduates and the short-term program graduates supported



by STEP over the project period). By the end of 2019, approximately 113,270 students²⁰ have or will have graduated from polytechnics supported by the STEP over the project period of the AF. Among them, around 41,600 polytechnic graduates have benefited from the stipend program. The pass rate (graduation rate) has improved from 64 percent in 2015 (the end of the original STEP project) to 86 percent in 2018. In this analysis, the impact of the IDG and the impact of stipend are estimated separately since the pass rates of stipend students (91% on average) are distinguishably higher than the overall pass rate (87% on average). Besides, it is reasonable to assume that the program impact of the stipend along with IDG on enrollment would be higher than of the IDG only.

51. Several scenarios were considered, varying both the enrollment increase of stipend beneficiaries and the wage premium for graduates of short-term programs. The lowest impact scenario (30% enrollment increase for the stipend beneficiaries and 10% wage premium for short-term programs) yields a Net Present Value (NPV) of US\$495 million and an Internal Rate of Return (IRR) of 18.7%, and the highest impact scenario (50% enrollment increase for the stipend beneficiaries and 30% wage premium for short-term programs) yields NPV of US\$576 million and an IRR of 19.2%. As a comparison, the re-evaluated NPV of the STEP is higher compared to the one at the Appraisal, mainly due to inclusion of benefits from the short-term programs and because the actual pass-rate of STEP-supported graduates (87%) is higher than assumed at Appraisal (high case of 70%). In summary, the Project's results justify its investment.
52. **The Project was also implemented efficiently.** Supported by a strong PIU, the Project that counted upon two additional financings was implemented over nine years. At completion, the PIU was staffed by about 50 staff members responsible for coordinating project activities among over 250 training institutions and four sector institutions. There was an initial delay in the utilization of IDGs by polytechnics as it was a new concept in TVET and required a substantial capacity development of polytechnics for fiduciary management. However, the delay was fully recovered afterward as the stakeholders gained the capacity to implement IDG activities. The PIU also intervened directly by centrally procuring some of the common items on the institutions' behalf. The total cost for project management amounted to US\$11 million over the nine years or approximately 5.2 percent of project costs. The Closing Date was extended only to allow additional time to complete activities incorporated under the second AF. Upon completion, 100% of the funds under the original credit, the first and second AF were disbursed (see Financing section of Data Sheet for disbursement ratios).

Assessment of Efficiency and Rating

53. Efficiency is rated **High** based on its higher than appraised estimated return, the reasonable costs of project management, and the high efficiency with which the Project was implemented, with original activities scaled up and new ones introduced through successive additional financings.

²⁰ 2016 and 2017 graduate numbers are 37,319 and 19,316, respectively. The 2018 and 2019 graduate numbers are estimated based on past two years graduate numbers.



D. JUSTIFICATION OF OVERALL OUTCOME RATING

54. Relevance is rated High, Efficacy is rated Substantial, and Efficiency is rated High, as described above. The Project's objectives have been achieved at a high level exceeding all the PDO Indicator targets, substantiated by evidence from Intermediate Indicators, other studies, and M&E report, with a minor caveat that one objective could not be fully measured by PDO Indicators. Hence, the Project's Overall Outcome Rating is **Satisfactory**.

E. OTHER OUTCOMES AND IMPACTS (IF ANY)

Gender

55. The enrollment of female students in TVET was particularly low at the time of Appraisal (5 percent). The Government's 7th Five Year Plan explicitly encourages female enrollment in TVET and aims at increasing the females' ratio in the overall TVET from 27 percent in 2015 to 40 percent by 2020²¹. By the time of the MTR, little progress had been made in terms of expanding female enrollment in polytechnics and ensuring their retention. Only 12 percent of the Project's stipend beneficiaries were female. Also, the high opportunity cost of studying for four years due to the availability of jobs in the RMG sector and other industries appeared to impact the high rate of drop-out. To help increase enrollment and reduce drop-out, the Project: (a) expanded the provision of stipends to all female students in eligible project-supported institutions; (b) prioritized removing the supply-side constraints by improving institutional environments and offering more female-favored training courses such as garments, computer, civil engineering and electronics; and (c) increased awareness-raising. As a result, the enrollment of females in TVET institutions increased to over 20 percent by completion. Women's participation in the program helped enhance employment opportunities and promote their empowerment.

Institutional Strengthening

56. In addition to the strengthening of TVET institutions, the Project had an essential role in supporting the development and enhancement of the institutional capacity of key government stakeholders responsible for TVET sector management, operation, and coordination. The main focus was on support to DTE and the BTEB under the MOE, the BMET under the MOEWOE, and also on the ISCs and the NSDC. The Project helped strengthen the institutional framework for TVET by financing: (a) strengthening of ISCs²² and establishment of the Dhaka Technical Teachers Training Institute for training of teachers of short-course programs; (b) improvements in physical and information system capacity at DTE, BTEB, and BMET; (c) improvements to the efficiency of existing systems; and (d) improvements to administrative and managerial capacity. A total of 635 managers and staff of DTE, BMET, and BTEB received managerial and leadership training, both nationally and internationally (against a target of 15 managers).

²¹ A study on female participation in skills development diagnosed the main causes of low female participation in TVET as: (a) supply constraints-lack of institutions nearby, non-conducive facility and faculty environment, and non-availability of female friendly courses; (b) reputational issues as the TVET sector was associated with the image of male domination and low-performers; (c) weak demand for training among potential female trainees due to lack of knowledge; and (d) household financial constraints.

²² The ISCs are an important part of the system-level capacity to promote industry collaboration. As of December 2019, 12 ISCs have been established. Different development partners are supporting ISCs. The Project aimed to support two ISCs – ICT ISC and Light Engineering ISC – through NSDC. Due to complications with coordination with NSDC, the Project could provide assistance to ICT ISC only.



Mobilizing Private Sector Financing

57. The Project promoted public-private sector partnerships not only for financing but for technical inputs as well. Incorporating the private sector in TVET programs was very much at the heart of its planned quality improvements. To ensure the relevance of TVET to the needs of industry, and to expand the provision of quality TVET, the Project's design contemplated strengthening and sustaining continued linkages with the private sector in most areas. It also benefited from private sector expertise and training delivery, although the exact cost of these contributions has not been calculated. First, the IMC of each participating institution counted upon significant representation from the private sector. These IMCs provided guidance on how the training institutions could adequately respond to the needs of the labor market and helped facilitate institutions' greater involvement with the industry by establishing 930 industry partners. Second, to level the playing field between public and private institutions, a deliberate decision was made at Appraisal that both public and private institutions would be allowed to compete for resources under the Project. Private institutions were eligible to apply for competitively funded IDGs to implement quality improvement activities, and students in these institutions were eligible to receive stipends. In all, 19 private polytechnics received funding for their IDGs under the Project. Also, 28 private training institutions (out of a total of 80 institutions) delivered short-term programs were supported by the Project. The Project also piloted public and private partnerships for enterprise-based training by private sector organizations. The Bangladesh Garment Manufacturers and Exports Association (BGMEA) trained 14,040 unskilled laborers in five of its training centers, providing food and accommodation support to disadvantaged trainees from rural areas with subsidized costs.

Poverty Reduction and Shared Prosperity

58. The Project aimed to improve access of TVET programs to marginalized communities through better targeting and more inclusive processes. The proxy means-tested (PMT) targeted stipend program was designed to ensure that persons from disadvantaged socio-economic backgrounds could participate in the TVET program, thus ensuring a policy of social inclusion. The Project provided stipends to 285,422 students in 162 polytechnics and 139,477 students in 71 short-term training programs. Of these, approximately 18 percent were female, while the others were from disadvantaged groups selected by PMT scores, or other qualifying conditions.²³

Other Unintended Outcomes and Impacts

59. The Project has helped improve the image of technical education in Bangladesh, with the result that enrollments are increasing rapidly, the educational standing of incoming students in terms of their grade point averages is increasing, and acceptance rates at selected institutes, especially the public polytechnics, are decreasing. Further, the Project helped host a show-case for the promotion of small-scale local innovation by TVET teachers and students. Teachers who were trained in Singapore benefitted from the experience and pursued small-scale research at their polytechnics. Students' innovations in TVET were promoted through the National Skill Competitions. Examples of students' innovations include Central Nebulizer & Suction Machine, Wheel Spray Pump, Digital Chicken Farming System, etc.

²³ This includes orphaned students, students from households with a divorced or widow headed family, or indigenous people.



III. KEY FACTORS THAT AFFECTED IMPLEMENTATION AND OUTCOME

A. KEY FACTORS DURING PREPARATION

60. **The STEP was responsive to the Government's policy priorities.** The GOB viewed the skilling of its human resources as critical for maintaining its competitive edge in global markets, and its focuses on education and training were expected to have a significant impact on the country's economic development. The Government's New Education Policy (NEP) 2010 set as a goal to increase the share of students in the TVET system to 25 percent by expanding access to quality vocation education. The NEP recognized the challenges that the TVET system was facing in linking the supply of skills to those in demand by the workplace and seek to promote greater private sector participation in the provision and determination of skill needs. At a broader level, the Government's priority was reflected in its Second National Strategy for Accelerated Poverty Reduction (NSAPR-II), a transitional poverty reduction strategy in place until the launch of its Sixth Five-Year Plan for FY11-15. All objectives were supported by the Project as an investment in quality tertiary and vocational education would be needed to produce better trained and educated workers required for improved productivity and higher-skilled jobs.
61. **The Project built upon assistance provided by other development partners, most notably the European Commission, International Labor Organization (EC-ILO), and the Asian Development Bank (ADB).** The EC-ILO Technical and Vocational Education and Training Reform Project was designed to support policy reforms of TVET, systems, and legislation, a new National Skills Development Policy (2011), National Technical and Vocational Qualifications, a new TVET Teacher Training Program, a new TVET Manager Training Program, Apprenticeship Guidelines and Guidelines for Recognition of Prior Learning. The ADB funded Skills Development Project, with similar objectives as the STEP, aimed to improve the relevance of and access to TVET programs and improve the capacity for quality TVET delivery. While there were no formal partnership arrangements, the STEP preparation team coordinated closely with teams of both the EC-ILO and ADB to ensure coordination in which the Project would build upon the policy reform efforts promoted by the EC-ILO project without duplication of efforts. This close coordination allowed the Project to focus the investment in institutional capacity building and expansion of training programs and maximize the return on investment.

B. KEY FACTORS DURING IMPLEMENTATION

62. **The economic growth and industry development scenario was consistently positive, creating a favorable environment for linkages with the private sector and demand for skills training.** Bangladesh maintained its robust growth performance, providing an environment conducive to the achievement of the Project's objectives. Bangladesh's economy grew, on average, 6.5 percent per annum, reaching 7.9 percent in 2017/2018. Exports and remittances have been buoyant. This rapid growth enabled Bangladesh to nurture a range of emerging industries and increase demand for various technical skills. Emerging industries such as ceramic, ICT, tourism, and construction have all become very active players in skills



development and essential partners for the government and institutions to upgrade skills training programs.

63. The Government provided sustained commitment to the PDOs throughout implementation.

Developing a quality skilled workforce to accelerate economic growth remained very much at the center of the Government's policy priorities over the project implementation period, and the Government took several important policy decisions that reflect this. Reforms envisaged in the NSDP include: (a) a shift to a new national skills qualification system, the National Technical and Vocational Qualification Framework (NTVQF), and (b) introduction of a series of new modes of skills development, apprenticeship and workplace training, Recognition of Prior Learning (RPL), and enhancing skills development for overseas employment. Then, to provide more focused priority to TVET within the MOE, in November 2016, the Government reorganized the ministry, creating a separate division responsible for Technical and Madrasah education. Finally, in 2018 legislation was passed transforming the NSDC²⁴ to the National Skills Development Authority, with full legal status as the sole authority for overseeing technical and vocational education and training in Bangladesh.

64. The performance of the Project Implementation Unit was strong with steady leadership.

The DTE, with the support of a Project Implementation Unit (PIU), was responsible for project implementation. The unit was responsible for all project fiduciary and safeguard processes for coordinating monitoring and evaluation processes and for periodic reporting, including coordinating fiduciary, safeguard, and monitoring processes among a vast number of participating institutions and providing stipends to their eligible students. Despite some initial delays in staffing critical positions in the PIU, its performance was reliable throughout implementation. Turnover was infrequent, and over the Project's nine years of implementation, the PIU was headed by only three Project Directors, each of which remained in the position for at least three years. The Project's implementation was well recorded and monitored, and its documentation provides a strong institutional memory.

IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME

A. QUALITY OF MONITORING AND EVALUATION (M&E)

M&E Design

65. The Project assigned high importance to M&E. The Project's M&E was designed to monitor project achievements, effective use of resources, promote learning from implementation experience, and measure project impact. M&E was to be coordinated by the M&E Unit, within the PIU, which would collect performance monitoring data from each of the participating institutions based on a format designed for the Project that included the Project's key performance indicators. The RF was concise, with regularly available data focused on those results that would be impacted by the Project's activities. However, the RF had a minor gap in measuring progress towards the employability of socio-economically disadvantaged students (Section III,

²⁴ The NSDC was the apex authority on skills development in Bangladesh, responsible for coordinating the provision of skills development, ensuring that public institutions are best utilized to maximize effective use of resources, and implementing arrangements so that private providers can access public facilities to increase the scope of skills training.



Objective III). The Project addressed this by providing gender-segregated data through its periodic monitoring reports. The M&E design also contemplated carrying out regular quality assurance surveys, annual performance audits, and evaluation studies. Also, the Project incorporated several activities aimed at improving the monitoring of training institutions, students, graduates through the development and implementation of management information systems in DTE, BTEB, and BMET (Section 1 A).

M&E Implementation

66. The implementation of M&E was slow to begin. At the start, the absence of an integrated system of data management and analysis, delays in systematic reporting, and weak utilization of the M&E data for timely decision-making constrained the use of evidence for improving training quality and delivery under the Project. The PIU addressed these issues systematically through staffing of the M&E Unit, and the design, development, and implementation of a functional management information system for the Project. Other TVET sector institutions counted upon the PIU's support to enhance their M&E capabilities, including the BTEB, BMET, and DTE, as well as participating in training institutions. Given the Project's increased scope and its corresponding requirements, staffing of the MIS Unit was reinforced after the second AF. Reporting was provided regularly in a timely fashion, and the RF was continuously updated with detailed information collected from beneficiary institutions. It was updated during each of the AFs and restructurings to reflect progress in implementation and additional activities that would be carried out with the additional funding. Further, a large number of assessments, surveys, tracer, and other studies (some carried out by the World Bank as technical assistance) were carried out to evaluate progress. These included: (i) two round of graduate tracer studies on short-course graduates; (ii) two rounds of graduate tracer studies on polytechnic graduates; (iii) effectiveness assessment on IDG support for polytechnics; (iv) evaluation of SSC (Voc) pilot program; (v) effectiveness study on stipend program; (vi) RCT-based RPL impact evaluation study; (vii) ex-post RPL effectiveness assessment; (viii) analysis on challenges for post-secondary TVET for female; (ix) stocktaking study of ISCs; (x) review on NTVQF; and (xi) teacher training effectiveness assessment.

M&E Utilization

67. Semi-annual progress reports were prepared in a timely fashion and submitted before each joint review mission. These reports helped the Government and World Bank implementation support teams identify progress and critical challenges and made the periodic support mission more effective. Project monitoring information and findings and recommendations of the assessments and surveys were used very effectively for introducing modifications in the Project's design in the AFs and restructuring (Section I B). In addition, the findings of the more recent assessments have been valuable for the design of a follow-on operation, currently under preparation (Section IV D).

Justification of Overall Rating of Quality of M&E

68. The Overall Quality of M&E is rated High. After addressing minor issues at the start of implementation, the Project's monitoring was timely, updated regularly, and provided thorough information, including gender-wise breakdowns. The Project's monitoring data, including that summarized in its RF, was used to guide implementation support. Together with the findings of the many surveys and evaluations carried out, the monitoring data helped feed into routine adjustments to the Project's design.



B. ENVIRONMENTAL, SOCIAL, AND FIDUCIARY COMPLIANCE

69. **Environmental Safeguards.** The Project triggered OP/BP 4.01 (Environmental Assessment) and was categorized under environmental Category B since it could finance rehabilitation/renovation/refurbishment of the existing buildings, classrooms, workshops, and library and enhancing workshops and laboratory facilities. While the environmental impacts of any possible works, the physical intervention and detailed design of the infrastructure were not known at Appraisal. Therefore, an Environmental Management Framework (EMF) was prepared to incorporate the guidelines necessary for the preparation of an Environmental Management Plan (EMP) for each relevant activity after conducting environmental screening. Throughout the project, the M&E Unit of the PIU conducted regular monitoring of environmental compliance and reported the findings. No significant issues were encountered.
70. **Social Safeguards.** The Project was expected to generate positive social impacts through its efforts to increase access of TVET programs to marginalized communities through better targeting and more inclusive processes. Nevertheless, this was a nation-wide program, and the exact locations of where activities would be financed had not been determined at Appraisal. Therefore, OP/BP 4.10 (Indigenous Peoples) was triggered if project activities would be implemented in areas that housed indigenous peoples and other vulnerable communities. A framework approach was adopted, and a Social Management Framework (SMF) was prepared. Throughout the project implementation, the M&E Unit of the PIU conducted regular monitoring of social safeguard compliance. No significant issues were encountered.
71. **Fiduciary compliance.** At Appraisal, risks for both procurement and financial management were considered High. This was mostly because of the large number of dispersed training institutions that would be procuring goods and services, and handling the related financial management under the framework of their IDGs. The Project financed the contracting of both procurement and financial management specialists for the PIU, and the PIU, in turn, worked with training institutions to provide hands-on support and training. Other than minor delays with staffing and then actual contracting and recommendations for documentation requirements, neither procurement or financial management presented significant problems. Procurement plans were updated regularly, and project audit reports were received in a timely fashion. For financial management, the Project was compliant in submitting the Interim Unaudited Financial Reports (IUFRs) and audit reports throughout the project implementation period, for both the Original and AF Credits. Except for a few minor exceptions on unreconciled balances of the utilization of funds between the designated account and client connection, the IUFRs were acceptable to the Bank. Also, to improve transparency and efficiency in procurement, the Project financed the introduction of e-Government procurement in DTE (and in the PIU), BTEB and BMET, and in polytechnics on a pilot basis. No significant issues were encountered.



C. BANK PERFORMANCE

Quality at Entry

72. **The World Bank's performance in ensuring quality at entry is rated Satisfactory.** The Project's preparation was grounded in the recommendations of a policy study and subsequent discussions with the Government. The preparation team helped put in place a project that responded directly to the Government's priorities for its TVET sector, and that was coordinated to complement assistance that was being provided by other donors (EC-ILO and ADB). The results framework was concise, with intermediate outcome indicators leading to the achievement of its PDO indicators. The team was focused on results, as reflected in the Project's planned implementation of several assessments and surveys throughout implementation. Institutional arrangements were complicated, in part reflecting the institutional complexity of Bangladesh's TVET sector. Still, the establishment and later training of a strong PIU with well-designed staffing arrangements and responsibilities within DTE were instrumental in ensuring smooth implementation and coordination among different training institutes and sectoral institutions. The Project was reasonable in terms of the activities it included and the objectives that it strived to achieve, within its given implementation period. Despite minor delays at the start of implementation, both the reasonableness of the Project's scope and the ultimate strong performance of the PIU ensured its early progress and the continued robust implementation through each of its AFs.

Quality of Supervision

73. The World Bank's quality of supervision is rated Satisfactory. Throughout implementation, the World Bank's implementation support team, led by Country Office-based staff, sustained a continuous dialogue with the PIU and other sector Institutions (BMET, BTEB) and training institutes. The team was pragmatic and results-focused and incorporated the findings of the many assessments and surveys effectively in the design of subsequent assistance under the AFs. The team also conducted periodic workshops with stakeholders of different project-supported programs to obtain feedback. Implementation progress, workshops' and studies' findings were used to introduce minor adjustments to project indicators and indicator targets and to feed into the preparation of a proposed follow-on project. The preparation and documentation of the AFs were timely, comprehensive, and responded to the Government's emerging priorities (e.g., RPL) in addition to lessons of implementation experience. The team helped forge formal training partnerships with different countries to the benefit of managers and teachers. The implementation missions' Aide Memoires were succinct and focused on both results and the next steps to advance implementation. The Project's Implementation Status and Results Reports were timely, transparent, and results-focused and provided a strong basis for understanding implementation issues and progress towards the Project's PDO.

Justification of Overall Rating of Bank Performance

74. Overall World Bank performance is rated Satisfactory based on Satisfactory ratings for both quality at entry and quality of supervision as described above.

D. RISK TO DEVELOPMENT OUTCOME

75. **The Risk to Development Outcome is considered Modest.** First and foremost, the Project's objectives continue to count upon the Government's highest priority. A well-functioning TVET sector, capable of delivering graduates with high-quality skills as required by the labor markets is fundamental for Bangladesh's sustained growth in the future. A mechanism for targeting



individuals from less advantaged backgrounds and females is also a Government priority, supported by its policy framework, and essential for helping further reduce poverty and ensuring that the benefits of growth are shared among all Bangladeshis. The Government has already internalized some of the Project's features, including a continuation of the stipend program with internal budget resources. The World Bank is assisting the Government in the preparation of a follow-on Accelerating and Strengthening Skills for Economic Transformation (ASSET) Project, for which proposed IDA financing is expected to be submitted for consideration by the Board of Directors in September 2020. Some challenges are anticipated, such as, among others, securing continued financing for the ever-changing training equipment requirements needed to maintain relevance to the constant introduction of new technologies.

V. LESSONS AND RECOMMENDATIONS

76. **Programs similar to Recognition of Prior Learning financed by the Project can have a positive impact on the many workers without formal education.** First, these programs work well in an environment of excess demand for skilled labor. Still, there is a strong case for budget financing of these programs as industries have few, if any, incentives to improve the qualifications of workers that are already performing tasks required by their trade. On the other hand, these programs work as a very targeted one-time (and one-cost) social protection initiative that aims to increase the qualifications and pay for many workers that possess skills required but are not compensated accordingly due to lack of formal training. Finally, while obtaining a certification of qualification improves workers' confidence and standing vis-à-vis colleagues and their community, the impact on workers' pay is not immediately apparent. Further, longer-term monitoring would be required to conclude the results.
77. **Continued communication is fundamental to alter public opinion on improvements to TVET, and Skills Competition is a highly effective intervention to improve social awareness on TVET.** Although the Project did make substantial progress in improving public perception of technical education, many people still have negative opinions on technical programs. The importance of a sustained effort to disseminate development and examples of success cannot be underestimated, especially when significant improvements have occurred to a previously under-performing system. The skills competition under the Project helped bring recognition to the TVET institutions and stimulated students to develop and document their innovations. This not only helps boost students' confidence but serves to promote TVET and attract the participation of industries and help strengthen linkages.
78. **The coordination of interventions and division of roles by separate donor-financed operations that support the common policy reform agenda is key to maximizing development impact for skills.** The Project was an Investment Project Finance that supported the implementation of essential policy innovations in the TVET sector. It did not contain specific requirements or activities aimed at enacting different policies, but only supported their implementation. The devising, defining, and implementing of policy innovations was supported by financing from the EC-ILO. With close coordination among the Government, the EC-ILO and the World Bank, this arrangement worked very fluidly and produced outstanding results, without overburdening the World Bank's investment loan.



79. **An innovative shared leadership structure of PIU allowed the Project to successfully involve multiple line ministries to implement well-aligned and coordinated skills interventions.** To accommodate the need for working across skills training institutions and agencies under different ministries, the Project devised an innovative way for harmonization of operation by placing multiple agencies under one PIU, which had a Project Director (PD) from MOE and Deputy Project Directors from relevant agencies. One shared designated account was used to finance all the project activities with the approval authority of the PD. This arrangement enabled the Project to minimize the management costs and maximize coordination and synergy across operations for different agencies, including synchronization of training curriculum standards across DTE/BTEB and BMET.
80. **Training institutions providing housing facilities need to create friendly environments to attract female students, and school leadership needs to support this.** There are often legitimate concerns among both women and their families in sending their daughters off to reside away from home. Schools that have been successful in attracting women have had support from the top to help attract women students and provide them with an accommodating atmosphere away from home. This could include, for instance, close circuit cameras, separate dormitories where they report feeling safe, separate leisure rooms and washrooms, and friendly relations with teachers, where females are strongly represented.



ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS

A. RESULTS INDICATORS

A.1 PDO Indicators

Objective/Outcome: Improving training quality of vocational training

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Pass rates of students from supported Diploma programs (%)	Percentage	50.00 01-Sep-2009	60.00 30-Jun-2016	65.00 30-Jun-2019	86.50 30-Jun-2019
Female	Percentage	66.00 30-Oct-2015	68.00 30-Jun-2019	68.00 30-Jun-2019	89.50 30-Jun-2019
Disadvantaged	Percentage	55.00 30-Oct-2015	60.00 30-Jun-2019	60.00 30-Jun-2019	92.50 30-Jun-2019
Comments (achievements against targets): National average:77.9%					



Objective/Outcome: Improving employability of trainees

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Employment rates for trainees of short-term courses within 6 months of course completion	Percentage	30.00 01-Sep-2009	44.00 30-Jun-2016	46.00 30-Jun-2019	51.00 30-Jun-2019

Comments (achievements against targets):

Objective/Outcome: Improving access to training for those from disadvantaged socio-economic backgrounds

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of disadvantaged students supported through stipends selected based on household information based selection criteria	Number	0.00 01-Sep-2009	160000.00 30-Jun-2016	175000.00 30-Jun-2019	240118.00 30-Jun-2019

Comments (achievements against targets):



A.2 Intermediate Results Indicators

Component: Component 1: Improve the Quality and Relevance of Training

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Direct project beneficiaries	Number	240000.00 19-Aug-2015	420000.00 30-Jun-2019		706847.00 30-Jun-2019
Female beneficiaries	Percentage	14.00	17.00		20.70

Comments (achievements against targets):

A new indicator introduced at the Additional Financing in 2015

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
% of utilization of Institutional Development Grants under AF	Percentage	0.00 30-Jun-2015	95.00 30-Jun-2019		96.60 30-Jun-2019

Comments (achievements against targets):



Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Component 1.1: Enrollment as a proportion of sanctioned intake capacity in supported courses	Percentage	55.00 25-Jan-2010	65.00 30-Jun-2016	95.00 30-Jun-2019	99.00 30-Jun-2019
Comments (achievements against targets):					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Component 1.1: Share of vacancies in sanctioned faculty posts in supported courses	Percentage	25.00 25-Mar-2010	10.00 30-Jun-2016	5.00 30-Jun-2019	3.00 28-Sep-2018
Comments (achievements against targets):					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Component 1.1: Transition rates of students from second to third semester in supported courses	Percentage	80.00 25-Mar-2010	90.00 30-Jun-2016	95.00 30-Jun-2019	98.00 30-Jun-2019



Comments (achievements against targets):

This indicator intends to measure retention of students in polytechnics.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of faculty in supported courses trained	Number	1500.00	6000.00		7980.00
		15-Apr-2015	30-Jun-2019		30-Jun-2019

Comments (achievements against targets):

A new indicator introduced at the Additional Financing in 2015

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Component 1.2: completion rates for trainees in supported short-term training courses	Percentage	50.00	70.00	95.00	96.00
		25-Mar-2010	30-Jun-2016	30-Jun-2019	30-Jun-2019

Comments (achievements against targets):

Indicator Name	Unit of	Baseline	Original Target	Formally Revised	Actual Achieved at
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	Measure			Target	Completion
Share of girl trainees in supported short-term training courses	Percentage	5.00	20.00	30.00	32.00
		25-Mar-2010	30-Jun-2016	30-Jun-2019	30-Jun-2019
Comments (achievements against targets):					

Component: Component 2: Pilots in TVET

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Component 2.1 Number of Industry Skills Councils functioning supported by the project	Number	0.00	2.00	7.00	6.00
		25-Mar-2010	30-Jun-2016	30-Jun-2019	30-Jun-2019
Comments (achievements against targets):					

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Component 2.2: Evaluation of SSC (Voc)	Text	0	Evaluation Completed	Evaluation Completed	Evaluation Completed
		25-Mar-2010	30-Jun-2016	30-Jun-2019	30-Jun-2019
Comments (achievements against targets):					



Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of individuals assessed for RPL in supported centers	Number	1900.00 15-Apr-2015	30000.00 30-Jun-2019		31804.00 30-Jun-2019
Comments (achievements against targets):					

Component: Component 3: Institutional Capacity Development

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of staff trained in Leadership and Management areas.	Number	22.00 15-Apr-2015	150.00 30-Jun-2019		635.00 30-Jun-2019
Comments (achievements against targets): Another intermediate results indicator for Component 3 "Establishment of Central Job and Information Portal in DTE" was achieved and subsequently removed from the results framework.					

Component: Component 4: Project Management, Communications, and Monitoring and Evaluation



Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Component 4.1: Awareness raising campaigns	Number	0.00	12.00	30.00	36.00
		25-Mar-2010	30-Jun-2016	30-Jun-2019	30-Jun-2019
Comments (achievements against targets):					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Satisfaction Survey (Student, Faculty, and Employers') for IDG recipient institutions	Text	Nil	Survey Completed		Survey Completed
		15-Apr-2015	30-Jun-2019		30-Jun-2019
Comments (achievements against targets):					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Component 4.2: Assessment Studies	Text	Nil	2nd tracer study on polytechnic completed; RPL impact evaluation completed		2nd tracer study on polytechnic completed; RPL impact evaluation completed
		15-Apr-2015	30-Jun-2019		30-Jun-2019



Comments (achievements against targets):



B. KEY OUTPUTS BY COMPONENT

Objective/Outcome 1 : Strengthen training institutions to improve quality	
Outcome Indicators	1. Pass rates of students from supported Diploma programs
Intermediate Results Indicators	1. % of utilization of Institutional Development Grants under the Additional Financing 3. Share of vacancies in sanctioned faculty posts in supported polytechnics (regular and contractual) 4. Transition rates of students from second to third semester in supported courses 5. Number of faculty in supported courses trained 6. Number of staff trained in Leadership and Management areas.
Key Outputs by Component (linked to the achievement of the Objective/Outcome 1)	1. Improved teaching and learning environment in polytechnics and short-course training institutions 2. Improved teaching skills and subject knowledge of teachers 3. Improved teacher deployment at institutions 4. improved teaching and learning materials
Objective/Outcome 2 : Strengthen training institutions to improve the employability of trainees	
Outcome Indicators	1. Employment rates for trainees of short-term courses within 6 months of course completion
Intermediate Results Indicators	1. Completion rates for trainees in supported short-term training courses 2. Number of Industry Skills Councils functioning supported by the project 3. Evaluation of the SSC (Voc) Pilot



	4. Satisfaction Survey (Student, Faculty, and Employers') for IDG recipient institutions 5. Assessment studies in TVET
Key Outputs by Component (linked to the achievement of the Objective/Outcome 2)	1. Enhanced curriculum and practical skills training in line with industry needs 2. Enhanced industry participation in institutional management of training institutions 3. Enhanced industry attachment and apprenticeship program 4. Enhanced job placement services for trainees 5. Enhanced employer feedback and graduate tracking processes
Objective/Outcome 3 : Improve quality of training and employability for trainees from a disadvantaged socio-economic background	
Outcome Indicators	1. Number of disadvantaged students supported through stipends selected based on household information based selection criteria 2. Pass rates of students from supported Diploma programs <u>for females and disadvantaged</u> 3. Employment rates for trainees of short-term courses within 6 months of course completion <u>for female</u>
Intermediate Results Indicators	1. Enrollment as a proportion of sanctioned intake capacity in supported courses (%) 2. Share of girl trainees in supported courses % 3. Number of individuals assessed for RPL in supported centers. 4. Number of awareness raising campaigns
Key Outputs by Component (linked to the achievement of the Objective/Outcome 2)	1. Increased enrollment of female students in training institutions 2. Increased enrollment of low-income students in training institutions 3. Increased access to RPL assessment for informal sector workers 4. Increased awareness among vulnerable communities about skills development opportunities



ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION

A. TASK TEAM MEMBERS

Name	Role
Preparation	
Venkatesh Sundararaman, Nalin Jena	Task Team Leader(s)
Pema Lhazom	Senior Operations Officer
Mokhlesur Rahman	ET Consultant
Subrata S. Dhar	Senior Education Specialist
Sangeeta Goyal	Education Economist
Syed Rashed Al-Zayed	Economist
Rajendra Dhoj Joshi	Senior Education Specialist
Marta E. Molaes-Halberg	Lead Counsel
Burhanuddin Ahmed	Sr. Financial Management Specialist
Marghoob Bin Hussein	Procurement Specialist
Muhammed Riadul Islam	Procurement Analyst
Shakil Ahmed Ferdausi	Environmental Specialist
Fabio Pittaluga	Senior Social Development Specialist
Sabah Moyeen	ET Consultant
Bertha Mburugu	Program Assistant
Nazma Sultana	Program Assistant
Tanusree Talukdar	Program Assistant
Sushila Rai	Team Assistant
Supervision/ICR	
Mokhlesur Rahman, Shiro Nakata	Task Team Leaders at ICR
Amit Dar, Venkatesh Sundararaman, Nalin Jena, Shinsaku Nomura	Former Task Team Leaders
Ishtiak Siddique	Senior Procurement Specialist



Zafrul Islam	Senior Procurement Specialist
Marghoob Bin Hussein	Senior Procurement Specialist
Mohammad Reaz Uddin Chowdhury	Senior Financial Management Specialist
Mohammed Atikuzzaman	Senior Financial Management Specialist
Arvind Prasad Mantha	Senior Financial Management Specialist
Syed Rashed Al-Zayed	Senior Economist
Tashmina Rahman	Education Specialist
Subrata S. Dhar	Senior Education Specialist
Sabah Moyeen	Senior Social Development Specialist
Fabio Pittaluga	Senior Social Development Specialist
Shakil Ahmed Ferdausi	Senior Social Development Specialist
Nadia Sharmin	Senior Environmental Specialist
Karthika Radhakrishnan	Operations Officer
Mahtab Alam	Program Assistant
Nazma Sultana	Program Assistant
Shourov Kumar Sharma	Team Assistant
Aftab Uddin Ahmad	Consultant
Mustahsin-Ul Aziz	Consultant
Ahamad Tanvirul Alam Chowdhury	Consultant
Syeda Kashfee Ahmed	Consultant
Suzana Nagele de Campos Abbott	ICR Main Author

B. STAFF TIME AND COST

Stage of Project Cycle	Staff Time and Cost	
	No. of staff weeks	US\$ (including travel and consultant costs)
Preparation		
FY06	1.200	5,337.45



FY07	5.200	26,723.73
FY08	6.542	38,839.70
FY09	54.293	131,598.73
FY10	68.233	252,400.78
FY11	4.960	21,793.71
Total	140.43	476,694.10
Supervision/ICR		
FY09	.325	1,466.44
FY10	.670	2,683.75
FY11	40.151	153,292.43
FY12	52.083	125,911.67
FY13	31.830	134,479.84
FY14	56.520	483,076.27
FY15	69.599	481,553.89
FY16	67.892	560,028.90
FY17	68.925	636,482.39
FY18	41.840	315,765.41
FY19	81.111	390,884.16
FY20	15.880	143,999.32
Total	526.83	3,429,624.47

**ANNEX 3. PROJECT COST BY COMPONENT**

Components	Amount at Approval (US\$M)	Actual at Project Closing (US\$M)	Percentage of Approval (US\$M)
Improve the Quality and Relevance of Training	162.2	155.65	95.96
Pilots in TVET	7.8	6.17	79.10
Institutional Capacity Development	40.8	34.75	85.17
Project Management, Communications, and Monitoring and Evaluation	14.7	13.00	88.44
Total	225.50	209.57²⁵	92.94

²⁵ Difference with the total disbursed amount in the data sheet is due to exchange rate differences.



ANNEX 4. EFFICIENCY ANALYSIS

The economic impact of STEP is estimated as the increased lifetime labor earnings of the graduates affected by the STEP. This increased lifetime labor earnings are estimated using the present discounted value (PDV) method. Three major benefits of STEP are generated from the following beneficiaries: (i) polytechnic graduates who wouldn't have graduated in the absence of the STEP's institutional grants for improving the quality of training (Sub-component 1.1); (ii) polytechnic graduates who wouldn't have graduated in the absence of stipends provided by STEP (Sub-component 1.1); and (iii) the employed graduates of short-term programs (Sub-component 1.3). The Present Discounted Value (PDV) of expected economic costs comprised both direct (project cost of STEP and students' out-of-pocket cost) and indirect (forgone labor earnings of the students) costs of completing the TVET diploma.

Economic analysis at the appraisal

The cost-benefit analysis of the STEP AF at the appraisal focused on the largest activity, Component 1.1, including the institutional development grant (IDG) and stipend support to polytechnics.

Benefits: Two main channels of benefits stream are quantified: (i) increased number of polytechnic graduates as a result of increased enrollment and increased pass rates, and (ii) increased wage premiums as a result of a better quality of training.

Costs: Given that the focus was on the Sub-component 1.1, the costs include the direct project cost of Sub-component 1.1 (\$US47 million), out-of-pocket expenses per polytechnic students, and indirect costs of foregone earnings due to the education.

Table 1. Basic Assumptions and Input Parameters

Description	Values used	Note
Sustained benefits of the institutional grant	10 years	Assumption - The benefits continue 10 years after the completion.
The annual growth rate of polytechnic enrollment	3 %	Assumption
Intake ratio with STEP (%)	95	Result indicator
Intake ratio without STEP (%)	70	Result indicator
Pass rate with STEP (%)	65	Result indicator
Pass rate without STEP (%)	55	Result indicator
Wage increase due to quality improvement (%)	5	
Active working years (years)	40	From age 20 to 60
Average start salary of TVET graduates (BDT)	10,077	ESS 2010; adjusted for the inflation from 2010 to 2015
Average start salary of SSC graduates (BDT)	7,687	ESS 2010; adjusted for the inflation from 2010 to 2015



Employment rate for TVET graduates (%)	95	Employment rate of TVET graduate to be 7% lower than that of SSC graduates (LFS 2010)
Employment rate for SSC graduates (%)	88	LFS 2010
Net annual out-of-pocket expense (the average total expense less the amount of stipend) (BDT)	3,690 BDT	13,290 - BDT800*12 months (HIES 2010)
Real interest rate (%)	8.1	The average of the past 20 years from 1994 to 2013 (World development indicators)
Exchange rate (BDT/US\$)	77.95	As of Jan 1, 2015 (Bangladesh Central Bank)

Result: The cost-benefit analysis using 9 different scenarios with the rates of graduation (pass rates) and the wage premium due to quality improvement was conducted. The scenario (65 percent pass rate and 3 percent wage premium) yields a net present value of US\$314 million with an internal rate of return of 12 per cent. NPV ranges between US\$137 million and US\$620 depending on the scenario.

Economic Re-evaluation

The cost-benefit analysis is evaluated using the outcome indicators (real pass rates and the number of polytechnic graduates and the short-term program graduates supported by STEP over the project period). By the end of 2019, approximately 113,270 students²⁶ have or will be graduated from polytechnics supported by the STEP over the project period of AF. Among them, approximately 41,600 polytechnic graduates have benefited from the stipend program. The pass rate (graduation rate) has improved from 64 percent in 2015 (the end of the original STEP project) to 86 percent in 2018.

In this analysis, the impact of the IDG and the impact of stipend are estimated separately since the pass rates of stipend students (91 percent, on average) are distinguishably higher than the overall pass rate (87 percent, on average). In addition, it is reasonable to assume that the program impact of the stipend along with institutional grants on enrollment would be higher than the one of the grants only.

In order to calculate the net impact of the IDG and the stipend, the counterfactual (the number of graduates in the absence of STEP) is estimated using the pass rate at the national level. Other input parameters that are used for the economic benefit of these sub-components are the following.

Table 2. Updated input parameters

Input Parameters	Value	Note
Sustained impact on pass rate	4 years	Assuming that the impact will be linearly depreciated over 4 years after the completion.
Intake ratio with STEP (%)	95	Value at appraisal
Intake ratio without STEP (%)	70	Value at appraisal
Enrollment increase (%) with	30% ~ 50%	Scenario base, empirically it is shown as high

²⁶ 2016 and 2017 graduate numbers are 37,319 and 19,316, respectively. The 2018 and 2019 graduate numbers are estimated based on past two years graduate numbers.



STEP stipend		as 65%
Pass rate with STEP (%), no stipend	87	Updated according to the result indicator
Pass rate with STEP (%), stipend	91	Updated according to the result indicator
Pass rate without STEP (%)	78	Pass rate at the national level
Wage Increase due to quality improvement (%) – TVET diploma	5	Base scenario value at the appraisal
Monthly Average Salary of TVET Graduate (unemployment adjusted), BDT	10492	Value at appraisal, adjusted using an annual wage increase of 3 % (LFS 2015-16 and 2017-18) from 2016 to 2019, an unemployment rate of 95%
Monthly Average Salary of SSC Graduate (unemployment rate adjusted), BDT	7414	Value at appraisal, adjusted using an annual wage increase of 3 % (LFS 2015-16 and 2017-18) from 2016 to 2019, an unemployment rate of 88%
Wage premium due to quality improvement (%) – Short term programs	10~30	Scenario base. Within 2 years, the wage increase for STEP supported short term programs graduate ranges between 27 and 34 %, in the meantime, the wage increase of non-STEP short-term program graduate is only 9% (Nomura et al. 2014). Assuming it lasts 5 years after completion.
Impact of STEP short course on employment	8	The difference in employment rates between the STEP supported and non-supported graduates is between 6% and 10% (Nomura et al.)
Real interest rate (%)	4	Average from 2015 to 2018
Exchange rate (BDT/US\$)	84.4	September, 2019

i) Economic benefit of IDG

Over the project period, the pass rate of students from the STEP-supported institutes was 9 percent higher than the national pass rate of polytechnic students. Along with the assumption of 25 percent higher enrollment due to the STEP, it is estimated that the STEP generated 11.5 percent more graduates over the project period. The economic benefit of the IDG is estimated by the increased earnings of these graduates (11.5 percent) over their economically active years (40 years). In addition, it is assumed that the impact of IDG on the pass rate will be depreciated over 4 years after project completion. Besides its impact on enrollment and pass rate, it is assumed that the quality improvement of institute results in a 5 percent increase in average wages of TVET graduates. In this analysis, the economic benefit of the stipend is separated from the overall impact of grants in the analysis. The present value of the economic benefit of institutional grants is estimated to be US\$161 million.



ii) Economic benefit of stipend

Over the project period, the pass rate of the stipend beneficiary students was 13 percent higher than the national pass rate of polytechnic students. In the case of the stipend, the impact on the enrollment (take-up) would be higher than the case of IDG. Empirically, the impact of the voucher (financial support) in vocational training is shown to be as high as 65 percent (Vocational Education in Kenya - a Randomized Evaluation, 3rd Working Paper, 2015). However, since the STEP supported only the partial cost of students' living expenses, its impact on the enrollment would be lower. In this analysis, three different scenarios of 30 percent, 40 percent and 50 percent increase in enrollment are considered. Combined with its impact on the pass rate, these scenario cases are translated into 47 percent, 58 percent, and 70 percent increases in the number of graduates resulting from the stipend. The economic benefit of the stipend is estimated by increased earnings of these affected stipend graduates over their economically active years (40 years). All other assumptions and input parameters are the same as the case of IDG. The present value of the economic benefit of the stipend is estimated to range between US\$155 million and US\$231 million.

iii) Economic benefit of short-term programs

A total of 35,106 graduates (45%) of short-term programs are identified as currently employed. Given that the short-term programs are targeted to improve the competency of industry-demanded skills, the economic benefit of these programs (wage premium) could be sustained in the short- to medium-term. In addition to the wage premium, an increase in the employment rate can be counted as a result of the STEP. According to the tracer study (Nomura et al. 2014), the difference in employment rates between the STEP-supported and non-supported graduates ranges between 6 percent and 10 percent. Under the assumption of wage premium of 10 percent to 30 percent for 5 years after project completion and with the life-long earnings of those graduates who would not have found employment without the STEP supported short-term program, the present values are estimated to be ranging between US\$187 (10 percent wage premium) and US\$232 million (30 percent wage premium).

Table 3. Summary - Total benefit of each component (Present Value, 2019 US\$)

Institutional grant	161,278,962
Stipend - 30% enrollment increase	155,814,770
Stipend- 40% enrollment increase	193,496,878
Stipend- 50% enrollment increase	231,065,859
Short term course -10% wage premium	187,685,913
Short term course -20% wage premium	210,119,197
Short term course -30% wage premium	232,552,481

iv) Economic cost of STEP

The cost includes present values of total project expenditures over the project period, and direct (total out of pocket students' expenditure over 4 years) and indirect costs (forgone earning during schooling years).



v) Results

The lowest impact scenario (30 percent enrollment increase for the stipend beneficiaries and 10 percent wage premium for short-term programs) yields an NPV of US\$495 million and an IRR of 18.7 percent, and the highest impact scenario (50 percent enrollment increase for the stipend beneficiaries and 30 percent wage premium for short-term programs) yields an NPV of US\$576 million and an IRR of 19.2 percent. The re-evaluated NPV of the STEP is higher compared to the one at the appraisal, mainly due to the inclusion of benefits from the short-term programs and the actual pass-rate of STEP supported graduates (87 percent) being higher than assumption used at appraisal (high case of 70 percent). In summary, the results justify the investment made through STEP project.

Table 4. Summary–Cost Benefit Analysis (NPV in millions 2019 US\$; IRR in %)²⁷

	Assumptions: increase in enrollment rate resulting from the stipend					
	30%		40%		50%	
	NPV	IRR	NPV	IRR	NPV	IRR
Wage premium for short term program: 10%	495	18.7	513	17.6	531	16.7
Wage premium for short term program: 20%	518	20.3	536	19.0	553	17.9
Wage premium for short term program: 30%	540	22.0	558	20.5	576	19.2

²⁷ The NPV is higher for the higher increase in enrollment rate scenario while the IRR is lower. This is due to cost stream is different across the scenario - faster increase in first 4 years of opportunity cost due to higher enrollment.



ANNEX 5. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS

Government of the People's Republic of Bangladesh
Skills and Training Enhancement Project
Directorate of Technical Education
Technical and Madrasah Education Division
Ministry of Education
Agargaon, Dhaka
Website: WWW.step-dte.gov.bd Email: step.dte@gmail.com

Memo No: STEP/Admin(1)/50.13 (PCR)/ 2018- ১০৭

Date: December 11, 2019

Subject: Borrower's Comments on the STEP ICR

Heartfelt thanks to the World Bank team for sharing the ICR of Skills and Training Enhancement Project (STEP). I am delighted to congratulate you for the excellent documentation of the background, inception, implementation, and achievements of the project as well as its remarkable impact on skills and Technical and Vocational Education and Training (TVET) sector of Bangladesh. I don't have much to add to what you have succinctly described in the well-structured ICR. I may take the privilege to briefly summarize the successes of STEP in the following paragraphs. However, I would like to thank the World Bank for their support in implementing this project successfully and making it a flagship project for the sector in Bangladesh.

The project has contributed a lot in developing the technical education institutions of the country. Providing implementation grants to the selected public and private polytechnic institutes and short term training providers helped them to develop their labs and training facilities which were in a dire need to upgradation. The project also equipped the institutions to operate at full capacity by hiring around 1,100 teachers. The GoB recognized the need for the continuation of smooth operation of the institutions and decided to recruit more 20,000 teachers for all the technical education institutions. The institutions including the polytechnics and the short term training providers needed capacity development. The efforts of the project in training the teachers, administrators and policy makers at home and abroad has had massive impacts in developing such capacity.

One of the most crucial interventions done by the project was the introduction of the universal female stipend for all the girls enrolling into the polytechnics. This coupled with the extended advocacy campaign has had a significant impact on increasing the enrollment of females at diploma-in-engineering level. The success of this activity has led to the government internalizing the activity in a much bigger scale, by providing stipends to the students of both public and private technical education institutions. Such efforts not only increased the access to technical education to many of the students but also increased their performance in class and laboratories as displayed by the increase in pass rate. The project has focused on the employability of trainees through strengthened industry-institution collaboration, job fairs and industry partnered training programs. STEP also piloted the apprenticeship program for SSC (vocation) students and has received positive feedback in favor of scaling it up. The project has also assessed about 40,000 participants under its RPL



program and showed that there is a massive demand for such certification program, especially for the informal sector.

The project has also developed the capacity of the key TVET players such as DTE, BMET and BTEB. The trainings provided under the project to the officials and the innovations introduced through the project has increased the efficiency of these institutions. The introduction of "Skills Competition" through the project has generated a wave of enthusiasm regarding skills education and innovation, which was previously non-existent. STEP has had a significant contribution in slowly changing the negative mindset of the people regarding technical education in the country.

Dear World Bank friends, we believe that the ICR captures the whole story of STEP with all of its dimensions and with these words I humbly endorse the ICR and its statements.

Finally, I would like to reiterate what I have said on many occasions when your supervision missions met us and the larger skills development community that the good initiative taken by STEP to transform the skills and TVET sector of Bangladesh through its myriad interventions must not end. The impetus and the momentum generated by the project with the support of the Bank should continue to expedite more reforms and changes in the sector so that the country's institutions attain a place of respectability among its peers in the world and contribute to create wealth and prosperity for the wellbeing of our mass people.

The skills development community strongly feels that a successor project in the sector is direly required to maximize the benefit of demographic dividend. At this circumstances, the proposal of a new project named Accelerating and Strengthening Skills for Economic Transformation (ASSET) with higher financial and technical contribution from the World Bank has made us extremely happy and we look forward eagerly to the fruition of our expectations.



(A B M Azad, ndc)

Project Directorate (Additional Secretary)
Skills and training Enhancement Project

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Senior Operations Officer & TTL
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Copy to:

1. Secretary, TMED, Ministry of Education, Bangladesh Secretariat, Dhaka
2. Director General, Directorate of Technical Education, F-4/B Agargaon, Dhaka-1207
3. Director General, BMET, Kakrail, Dhaka.
4. Chairman, BTEB, Agargaon, Dhaka.
5. Office Copy/Master Copy.



ANNEX 6. SUPPORTING DOCUMENTS (IF ANY)

World Bank, Country Assistance Strategy for the People's Republic of Bangladesh for the Period FY11-14, Report No. 54615-BD dated July 30, 2010.

World Bank, Country Partnership Framework for Bangladesh for the Period FY16-FY20, Report No. 103723-BD dated March 8, 2016.

World Bank, Project Appraisal Document, Skills and Training Enhancement Project, Report No. 53549-BD dated May 11, 2010.

World Bank, Employability of Post-Secondary TVET in Bangladesh, Tracking Survey of Graduates of Polytechnics, Report No. AUS0000063, dated January 2, 2019.

World Bank, How Does the Short-Term Training Program Contribute to Skills Development in Bangladesh? A Tracer Study of the Short-Term Training Graduates, Report No. 82 dated September 2015, and Center for Research and Management Counseling, SRG Bangladesh Limited, Bangladesh: Skill and Training Enhancement Project (STEP).

World Bank, Bangladesh, Vocational Training Project, P009393, Staff Appraisal Report No. 2316a-BD, dated May 3, 1979 and World Bank, Bangladesh, Technical Education Project, P009440, Staff Appraisal Report No. 4852-BD, dated April 25, 1984.

World Bank, Learning for Job Opportunities: An Assessment of Vocational Education and Training in Bangladesh, Bangladesh Development Series Paper No 19, Report No. 45189 dated June 2007.

World Bank, Breaking the Glass Ceiling: Challenges to Female Participation in Technical Diploma Education in Bangladesh, Report No. AUS0000002 dated January 2017.

World Bank, Employability of Post-Secondary TVET in Bangladesh: Tracking Survey of Graduates of Polytechnics, Report No. AUS0000063 dated January 2018.

World Bank, Implementation Status and Results Reports and Mission Aide-Memoires.

Skills and Training Enhancement Project (STEP) Project Completion Report, dated June 2019.

Tracer Study of Graduates of Short Course Training Programs (2nd Round), dated May 24, 2018.

Report on Recognition of Prior Learning (RPL) Assessment, dated August 2016.

Evaluation of SSC (Voc) Apprenticeship Piloting Program dated July, 2018.

Ahmed, Shaikh S., Stipend Effectiveness of the Skills and Training Education Project (STEP), July, 2015.

Assessment of Stipend Program of the Skills and Training Enhancement Project, April 2019.



ANNEX 7. SUMMARY OF BORROWER ICR

PROJECT COMPLETION REPORT : IMED 04/2003 (Revised)

A. PROJECT DESCRIPTION :

- 01. Name of the Project** : Skills and Training Enhancement Project (STEP).
- 02. Administrative Ministry/Division** : Technical and Madrasah Education Division
81. under Ministry of Education.
- 03. Executing Agency** : Directorate of Technical Education (DTE).
- 04. Location of the Project** : The geographical coverage of the project is
Country
82. wide.
- 05. Objective of the Project** :
- The project development objective is to strengthen selected public and private training institutions to improve training quality, and employability of trainees, including those from disadvantaged socio-economic backgrounds.
- The specific objectives of the project are to:
- i) To enhance quality and relevance of technical and vocational education and training as a part of meeting the strategic options of the poverty reduction agenda of the government;
 - ii) To strengthen the overall Technical and Vocational Education and Training (TVET) system through (a) direct start-up and operational support to Industry Skills Councils (ISC) and the National Skills Development Council and (b) support to SSC (Vocational) schools;
 - iii) To strengthen the capacity of key institutions (DTE, BTEB and BMET) which play important role in the management and quality assurance of TVET sector; and
 - iv) To establish a project management and implementation structure, implement communication strategy and undertake monitoring and evaluation.

It is expected that on successful implementation, the project will result in with improving the quality; relevance and efficiency of the TVET education and training in Bangladesh with increased employability of the graduates coming out of the system which will ultimately help reduce poverty and improve the quality of life of the general mass in particular.

D. ACHIEVEMENT OF OBJECTIVES OF THE PROJECT:

Objectives as per PP	Actual achievement
i) To enhance quality and relevance of technical and vocational education and training as a part of meeting the strategic options of the poverty reduction agenda of the government;	-Enrollment has been increased from 55% to 99% as a proportion of sanction intake capacity 83. -Provide stipend to 711,000 student years for all female student and disadvantage male students - Percentage of pass rate achieving from 50% to 86.46% 84. -75% attendance must be needed for stipend 85. - Transition rate of students from second to third semester increased from 80% to 98%



	86. -BDT 33994.11 lakh will be disbursed to 56 polytechnic institutes for strengthened with modern equipment, faculty training, industry visit etc.- 87. - Competition between public and private TVET institutes has been established 88. -trained 139477 unemployed youth through short term training with free of cost 89. - completion rate has been increased from 50% to 96% regarding short term training (6 months/360 hours) 90. -Share of girls trainees will be increased from 5% to 30% regarding short term training (6 months/360 hours) 91. -Employment rate has been increased from 30% to 49% regarding short term training (6 months/360 hours) - Trained 138600 unskilled and semi-skilled labor through short term training (1 month) with free of cost and employment rate was 73.16%
ii) To strengthen the overall Technical and Vocational Education and Training (TVET) system through (a) direct start-up and operational support to Industry Skills Councils (ISC) and the National Skills Development Council and (b) support to SSC (Vocational) Schools,	-Project Provided seed financing and operational cost to NSDC and provided some equipment to the ISCs through NSDC - Project provided support to strengthening and established of ICT ISC -Strengthening Secondary School Certificate (vocational) 92. - Apprenticeships piloting for SSC (voc) implemented in 20 institutes to 9125 students 93.
iii) To strengthen the capacity of key institutions (DTE, BTEB and BMET) which play important role in the management and quality assurance of TVET sector; and 94.	95. 96. -Utilized BDT 26370.06 lakh for development of the key TVET organizations (DTE, BTEB, & BMET) 97. -Web-based MIS developed for DTE, BETB and BMET 98. -Provide contractual teacher (Full teaching strength) to the public institutes and reduced vacant post from 71% to 3% 99. -xxxxx Numbers of faculty trained through local training 100. -xxxxx Numbers of staff trained in core competencies in DTE, BTEB and BMET through partnership with different renowned institutes in the world 101. -xxxxxx candidate certified under RPL system - Project established a technical teachers training Institute at BK-TTC for enhanced training capacity of short term training provider - Provided support for vertical expansion of 1 story of BMET & 4-storied of DTE building 102. -
iv) To establish a project management and	Project staff and contractual teacher appointed by the Project



implementation structure, implement communication strategy and undertake monitoring and evaluation.	-Consultants are selected according to PPR 2008 Project provide stipend to the selected student -Project provide institutional grants to the selected institutes -Provide training to teachers, other officials & staff of key TVET institutes -Capacity of DTE, BTEB BMET and NSDC strengthen - Skill Competition (i. Inaugural ii. Institutional, iii. Regional and iv. National) held 4 times since 2014 (each time around 5500 student participated). --Procurement done in accordance with GOB/World Bank guidelines - Mass people aware about TVET & STEP - Digital contents and contents management system developed and distribution - Strengthen M&E activities
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4.4 Impact of the project- This is a social sector project where benefits of projects cannot be quantified in the real sense of the term. Quantification of benefits in financial terms is not only a complex exercise, its outcome is also not easily understood. It does not give any clear picture of the benefits the project may accrue. However, project try to articulate the direct and indirect impact of the project.

4.4.1 Direct: Direct impact of the project interventions will be measured through an impact study at a later stage. However, to measure immediate impact or result achieved the Semi-Annual Monitoring Reports and various study reports are relevant. Since inception significant progress was made by the project based on different key indicators. The direct impacts are given below as per the Revised Project Result Framework:

- In base year pass rates of students from supported diploma programs were 50% (Overall). In the year 2018 overall target was 65%; female 68% and disadvantage (Stipend students) were 60% as of 2018 actual achievement were overall 86.46%; female 89.48% and disadvantage (Stipend students)92.52%. Whereas, In 2018 National average pass rate was 77.92%.
- Employment Rates for trainees of short-term courses within 6 months of course completion was estimated at 30% percent in baseline, the achievement was 49% in 2018, while the project end target was 46%;
- Number of disadvantaged students supported through stipends selected based on household information based selection criteria was 240118 compared to end target was 175,000



- The targeted total number of beneficiaries and the share of female were estimated to 420,000 and 17% respectively in YR7 (2018). The achievement was 665,180 in 2018, of which the share of female students was 20.29%.
- Enrollment has been increased from 55% to 99% as a proportion of sanction intake capacity
- Transition rate of students from second to third semester increased from 80% to 98%
- Competition between public and private TVET institutes has been established
- In base year the Employment Rates for trainees of short-term courses within 6 months of course completion was 30%. As of 2018 actual achievement was 49% compared to target 46%;
- project providing 1100 contractual teacher to the 50 (49 of DTE and 1 for BMET) polytechnics institutes offering diploma level training programs to ensure full teaching strength by reducing the vacant posts from 71% to 3%.
- completion rate has been increased from 50% to 96% regarding short term training (6 months/360 hours)
- Share of girls trainees will be increased from 5% to 30% regarding short term training (6 months/360 hours)
- trained 139477 unskilled and semi-skilled labor through short term training (1 month) with free of cost and employment rate was 73.16%
- Since 2014 project organized national skills Competition to showcase and promote TVET. The event created a much-needed promotion of TVET and awareness-building among stakeholders and general public.
- STEP provided web based MIS including necessary hardware, software and network for key TVET organizations
- STEP provide workplace experience supports quality training and employment prospects for 9125 SSC (Voc) students.
- Since 2014 STEP Organized national skills Competition to showcase and promote TVET
- Project create a pool of NTVQF certified assessor up to Level IV for strengthened the BTEB and the overall TVET sector's capacity as a whole to deliver skills training, assessment and certification under the NTVQF framework.
- STEP conducted RPL assessment through 20 institutes. The RPL program has been a huge success with over 31,840 candidates assessed, 24,620 certified competent and an overall achievement rate of around 77 percent.
- The project supported the establishment of Institutional Management Committees (IMC) in all STEP-supported polytechnics and short course training providers to review the progress of the training provider and advise on ensuring the relevance of the training to local industry demands.



- The project has supported the establishment of employment cells in all beneficiary polytechnics and short course for expanding access to employment
- STEP evaluated the RPL assessment and found that RPL assessment creates the opportunities of skills workers to participate in higher level vocation and technical training and education as well as upscale the work force. Considering the comparison before and after RPL assessment for nature and type of job of competent and partially competent assesses as well as control respondents, the significant increase was observed for both competent and partially competent assess in respect of full-time and part time wage employment in public and private entity. From the chi squared value it was revealed that there was significant difference found between before and after RPL assessment in terms of responses on the job title/position. Most of the competent (77%) and partially competent (91%) assesses reported that they were benefitted through social recognition. Majority of the competent (63%) assesses reported that the real benefit was the increase of opportunity for higher level of RPL and 73% partially competent assesses reported that real benefit was recognition by colleagues and friends. The majority 42% competent and 43% partially competent assesses were interested to participate further in RPL assessment without financial assistance.
- We evaluated the SSC (VoC) apprenticeship program and found that, out of 700 treatment group respondents, 85.9% (593) students reported that the apprenticeship attachment training increased their skills in expertise; 80.1% (553) students stated that industry training increased their self confidence; 39.3% (271) students reported that they achieved the BTEB/NTVQF Competency Certificate through assessment; 19.7% (136) students reported that the apprenticeship pilot program created opportunity to get better job; 12.0% (83) students reported that the industry training increased the opportunity for higher education. In terms of benefits of BTEB/NTVQF Competency Certificate, out of 435 students who achieved competency certificates, 65.2% of them (283) reported that certificate created the opportunity to get better job; 41.2% (179) students stated it increased social image; 40.1% (174) students said it created opportunity to get higher income in the job; 31.6% (137) students opined that it created opportunity for higher study; and 8.5% (37) students reported that the self-satisfaction was achieved through apprenticeship assessment. Among 435 treatment group students who reported about the benefits of certificate, 91.8% (393) students highly confident on skills achieved through BTEB/NTVQF assessment and certification program; while 8.2% (32) students were confident.
- Benefit Cost Ratio (1:10.33) of apprenticeship pilot program evidently revealed that the return against investment was very high. This return will be continued for the STEP supported graduate throughout his/her working life and that will be gradually increased through his/her life.